

# A Formally Verified Pandrosion–Steffensen Scheme

Closed-form real-line analysis and complex multi-start extension  
for the equation  $z^p = x$ , with a Lean 4 / Mathlib v4.7.0 development

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[github.com/ivan-fr/poussiere\\_de\\_x](https://github.com/ivan-fr/poussiere_de_x)

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## Abstract

This paper combines, in a single document, the real-line analysis of Pandrosion’s classical geometric construction for  $p$ -th roots and its formally verified complex-multi-start extension.

**Part I (real line).** We revisit Pandrosion’s construction reported by Pappus for approximating  $\sqrt[3]{2}$  and generalize it to the computation of  $p$ -th roots of any positive real  $x$  by inserting proportional means via Thales’ theorem. After normalization the iteration reduces to the universal form  $s_{n+1} = 1 - (x - 1)/(x S_p(s_n))$  with  $S_p(s) = 1 + s + \cdots + s^{p-1}$ , fixed point  $s^* = x^{-1/p}$ , and readout  $v_n = x s_n^{p-1} \rightarrow x^{1/p}$ . We derive the closed form

$$\lambda_{p,x} = \frac{(\alpha - 1) \sum_{k=1}^{p-1} k \alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k}, \quad \alpha = x^{1/p},$$

for the linear contraction ratio, prove it is monotone in both  $p$  and  $x$  uniformly (via the closed-form identities  $P_p(\alpha) = p D'_p(\alpha)$  and  $B_p(\alpha) = \sum_{j=1}^p j \alpha^{j-1}$ , both with all positive coefficients), with explicit large- $p$  limit  $1 - \ln(x)/(x - 1)$ , give non-asymptotic error bounds, extend the theory to  $x < 1$  with an explicit threshold  $\alpha_p^*$  below which the iteration diverges, and add two accelerations: a canonical starting point  $s_0 = h(1) = 1 - (x - 1)/(xp)$ , and Aitken–Steffensen  $\Delta^2$  yielding quadratic convergence with a much smaller constant than Newton. A scaling preconditioning  $x = A \cdot x'$  exploits the monotonicity of  $\lambda$  to collapse iteration counts for large  $x$ .

**Part II (complex extension, formally verified).** We extend the iterator  $h_{p,x}$  to  $\mathbb{C}$ , where it has  $p$  attractive fixed points — the  $p$ -th roots of  $1/x$  — separated by a non-trivial Julia structure. A multi-start scheme  $\text{seed}_{\alpha,\varepsilon}(z) = \alpha + \varepsilon(z - \alpha)$  with target-dependent  $\varepsilon$  places any  $z \in \mathbb{C}$  inside the quadratic-contraction disk of a chosen anchor  $\alpha$ , providing a universal convergence theorem with closed-form iteration count  $N_{\text{eff}} \sim \log_2 \log(1/\varepsilon)$ . A compatibility bundle (Theorem 16.4) collects target selection, Steffensen fixing, scaling compatibility, reconstruction, and optimal seed into a single glue statement that downstream modules cite by name. Unconditionally (without seeding), each anchor’s basin of attraction contains an explicit disk of radius  $R_\sigma(x, p, \gamma_k)$ , giving the lower bound  $\text{vol}(\bigcup_k \Omega_k) \geq \sum_k \pi R_\sigma(x, p, \gamma_k)^2$  on the total basin volume (Proposition 16.1). The induced solution map is Borel-measurable everywhere and continuous outside the Voronoï boundary of the cyclotomic anchor set (which is Lebesgue-null). All of Part II is formally verified in a Lean 4 / Mathlib v4.7.0 development of 121 modules with zero `sorry` and audited axiomatic closure reducing to the standard Lean whitelist.

The paper makes no claim of progress on classical open problems of polynomial root-finding; its main value is to record a concrete derivative-free root-finding scheme with closed-form convergence analysis and complete formal verification for the monomial family  $z^p = x$ .

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## Part I

# Real-line theory

## 1 Introduction

The Greek problem of doubling the cube consists of constructing a segment of length  $\sqrt[3]{2}$  from a unit segment. It is equivalent to finding two proportional means  $m_1, m_2$  between 1 and 2, that is,

$$\frac{1}{m_1} = \frac{m_1}{m_2} = \frac{m_2}{2},$$

which yields  $m_1 = \sqrt[3]{2}$  and  $m_2 = \sqrt[3]{4}$ . Pandrosion of Alexandria proposed an *approximate* geometric method, reported by Pappus in the *Collectio*, Book III [3]: by successive parallels (Thales' theorem), she constructs proportional ratios that approach these means without ever reaching them exactly in a finite number of steps.

This work starts from a fundamental observation:

*Mathematically the target number is exact, but any approximation procedure yields a measurable residual profile — a footprint intrinsic to the chosen method.*

At each finite step of Pandrosion's construction, there remains a gap  $\varepsilon_n = |v_n - \alpha|$  between the geometric approximation and the ideal value. The sequence  $(\varepsilon_n)$  is the **residual profile** of the method. Our goal is to compute this profile exactly for any  $p$ -th root  $x^{1/p}$ , and to identify the contraction ratio that characterizes Pandrosion's method.

**Positioning and related work.** Rational iterations for  $p$ -th roots are a classical topic with a large literature: Newton's method applied to  $f(u) = u^p - x$ , Halley's method, and more generally Schröder's (1870) and König's (1884) families produce iterations of arbitrary order with explicit asymptotic constants. The systematic analysis of fixed-point iterations and their acceleration goes back to Ostrowski [9], Traub [11], and Householder [10]; the derivative-free counterpart via Aitken–Steffensen (1926/1933) is standard and appears in every modern numerical-analysis textbook [5, 6]. What is specific to the iteration (3) below is that it arises directly from Pandrosion's geometric construction, *without* being derived as Newton or Schröder applied to a particular polynomial: it is a rational iteration with linear (not quadratic) order, whose contraction ratio admits the closed form (5) as a rational function of  $\alpha = x^{1/p}$ . We have not located this closed form, together with its monotonicity in  $(p, x)$  and its use to drive a scaling preconditioning, in the textbooks [5, 6] or the historical literature on cube duplication, but we make no claim that the iteration itself is genuinely new: the contribution is to record the profile of a specific historical iteration and to show how standard acceleration devices (Aitken  $\Delta^2$ , scaling) interact with it.

**Outline.** Section 2 recalls Pandrosion's construction for  $\sqrt[3]{2}$  and analyzes its residual profile. Section 4 states the generalization to any  $p, x$ . Section 5 computes the contraction ratio  $\lambda_{p,x}$  and provides closed-form expressions. Section 7 presents numerical tables. Section 8 compares with bisection, the secant method, and Newton's method. Section 9 establishes functional properties of  $\lambda_{p,x}$ . Section 10 provides non-asymptotic error bounds. Section 11 extends the theory to  $x < 1$ . Section 12 determines the optimal starting point. Section 13 shows that Steffensen's acceleration of Pandrosion achieves quadratic convergence. Section 14 exploits the monotonicity of  $\lambda_{p,x}$  to optimize the method for large  $x$  via scaling. Section 22 concludes.

## 2 Pandrosion's construction for $\sqrt[3]{2}$

### 2.1 The two sequences

The geometric construction produces two sequences as a function of an initial parameter  $u_0$ :

$$u_{n+1} = \frac{u_n}{2 - 2\left(\frac{4 - u_n}{4}\right)^3}, \quad v_n = 2\left(\frac{4 - u_n}{4}\right)^2, \quad u_0 = 0.5. \quad (1)$$

The sequence  $(u_n)$  is the *internal parameter* of the construction (a length in the Thales diagram). The sequence  $(v_n)$  is the *readout*: the approximation of  $\sqrt[3]{2}$  produced at step  $n$ .

**Proposition 2.1.**  $\lim_{n \rightarrow \infty} v_n = \sqrt[3]{2}$ .

### 2.2 Normalized substitution

Setting  $s_n = (4 - u_n)/4$ , so that  $s_n \in (0, 1)$ , we obtain:

- $u_n = 4(1 - s_n)$ ,
- $v_n = 2s_n^2$ ,
- $u_{n+1} = \frac{4(1 - s_n)}{2(1 - s_n^3)} = \frac{2}{1 + s_n + s_n^2}$ .

The iteration on  $s_n$  then becomes:

$$s_{n+1} = 1 - \frac{u_{n+1}}{4} = \frac{2s_n^2 + 2s_n + 1}{2s_n^2 + 2s_n + 2}. \quad (2)$$

### 2.3 Fixed point

The fixed point  $s^*$  satisfies  $s^* = (2s^{*2} + 2s^* + 1)/(2s^{*2} + 2s^* + 2)$ , which gives  $2(1 - s^*)(1 + s^* + s^{*2}) = 1$ , i.e.  $2(1 - s^{*3}) = 1$ , hence  $s^{*3} = 1/2$ , so:

$$s^* = 2^{-1/3} = \frac{1}{\sqrt[3]{2}}.$$

The output at the fixed point is  $v^* = 2s^{*2} = 2 \cdot 2^{-2/3} = 2^{1/3} = \sqrt[3]{2}$ . ✓

### 2.4 Residual profile: linear convergence

Let  $h(s) = (2s^2 + 2s + 1)/(2s^2 + 2s + 2)$ . The derivative at  $s^*$  gives the convergence rate:

$$h'(s^*) = \frac{4s^* + 2}{(2s^{*2} + 2s^* + 2)^2} = \frac{2(2s^* + 1)}{4(s^{*2} + s^* + 1)^2} = \frac{2s^* + 1}{2(s^{*2} + s^* + 1)^2}.$$

Substituting  $s^* = 2^{-1/3}$  and using the closed form (7):

$$\lambda_{3,2} = \frac{(\sqrt[3]{2} - 1)(\sqrt[3]{2} + 2)}{\sqrt[3]{4} + \sqrt[3]{2} + 1} \approx 0.2202.$$

**Theorem 2.1** (Linear residual law for  $p = 3$ ,  $x = 2$ ). Under the above hypotheses, for any  $s_0$  sufficiently close to  $s^*$ :

$$\varepsilon_{n+1}^{(s)} \approx \lambda_{3,2} \varepsilon_n^{(s)}, \quad \lambda_{3,2} \approx 0.2202.$$

For the output, a Taylor expansion of  $v(s) = xs^{p-1}$  at  $s^* = \alpha^{-1}$  gives  $v_n - \alpha \approx x(p - 1)(s^*)^{p-2}(s_n - s^*)$ . Using  $x(s^*)^{p-2} = \alpha^p \cdot \alpha^{-(p-2)} = \alpha^2$ , this rewrites as  $v_n - \alpha \approx (p - 1)\alpha^2(s_n - s^*)$ ; the two forms are equal. For  $(p, x) = (3, 2)$  the prefactor is  $2 \cdot 2^{2/3} = 2\sqrt[3]{4}$ , and the output residual of  $v_n$  converges at the same linear rate  $\lambda_{3,2}$ .

| $n$      | $s_n$      | $v_n$         | $\varepsilon_n =  v_n - \sqrt[3]{2} $ | $\varepsilon_{n+1}/\varepsilon_n$ |
|----------|------------|---------------|---------------------------------------|-----------------------------------|
| 0        | 0.8750     | 1.5312        | $2.713 \times 10^{-1}$                | —                                 |
| 1        | 0.8107     | 1.3143        | $5.44 \times 10^{-2}$                 | 0.200                             |
| 2        | 0.7974     | 1.2717        | $1.17 \times 10^{-2}$                 | 0.216                             |
| 3        | 0.7945     | 1.2625        | $2.58 \times 10^{-3}$                 | 0.219                             |
| 4        | 0.7939     | 1.2605        | $5.67 \times 10^{-4}$                 | 0.220                             |
| $\infty$ | $2^{-1/3}$ | $\sqrt[3]{2}$ | 0                                     | $\rightarrow \lambda_{3,2}$       |

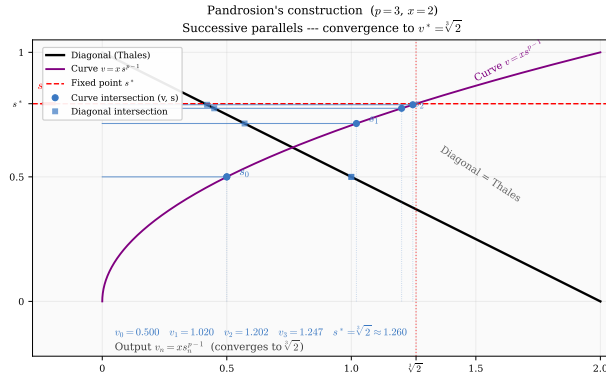
**Table 1:** Numerical residual profile for  $p = 3$ ,  $x = 2$  ( $u_0 = 0.5$ , i.e.  $s_0 = 0.875$ ). The ratio converges to  $\lambda_{3,2} \approx 0.220$ .

### 3 Geometric origin: why $S_p$ arises naturally

#### 3.1 The construction by successive parallels

Pandrosion's construction relies on Thales' theorem applied to a rectangle. Consider a rectangle with sides  $H$  and  $W$  equipped with two diagonals (Figure 1). A horizontal line at height  $u_n$  from the bottom intersects the two diagonals and defines:

- the parameter  $s_n = (H - u_n)/H$  (normalized position from the top),
- the readout length  $v_n = x \cdot s_n^{p-1}$  (the output of the construction).



**Figure 1:** Pandrosion's geometric construction for  $p = 3$ ,  $x = 2$ . The blue horizontal lines (steps  $n = 0, 1, 2, 3$ ) converge toward the dashed red line, which corresponds to  $v^* = \sqrt[3]{2}$ .

#### 3.2 Where does the polynomial $S_p$ come from?

The fundamental idea is as follows. We seek to insert  $p - 1$  proportional means between 1 and  $x$ , that is, to construct the ideal geometric sequence  $1, \alpha, \alpha^2, \dots, \alpha^{p-1}, x$  where  $\alpha = x^{1/p}$ . If  $s_n \approx \alpha^{-1}$  is the current approximation of the inverse ratio, the sum of the approximate geometric sequence is:

$$S_p(s_n) = 1 + s_n + s_n^2 + \dots + s_n^{p-1}.$$

The ideal sequence satisfies  $x \cdot S_p(s^*) \cdot (1 - s^*) = x - 1$ , which rewrites as  $x(1 - s^{*p}) = x - 1$ . Pandrosion's correction adjusts  $s_n$  so as to reduce the gap between  $S_p(s_n)$  and its ideal value via the formula:

$$s_{n+1} = 1 - \frac{x - 1}{x \cdot S_p(s_n)}.$$

Geometrically, the ratio  $(x - 1)/(x \cdot S_p(s_n))$  measures the *deviation* of the current proportional chain from the perfect geometric chain, and subtraction from 1 produces the corrected ratio. The polynomial  $S_p$  is thus the analytical transcription of the sum of the  $p$  segments constructed via Thales' theorem.

## 4 Generalization: Pandrosion's construction for $x^{1/p}$

### 4.1 The universal iteration

Let  $x > 0$  and  $p \geq 2$  be an integer. Set  $\alpha = x^{1/p}$ . The generalized Pandrosion geometric construction is defined as follows.

**Definition 4.1** (Generalized Pandrosion iteration). We define the sequence  $(s_n)_{n \geq 0}$  in  $(0, 1)$  by

$$s_{n+1} = 1 - \frac{x - 1}{x \cdot S_p(s_n)}, \quad S_p(s) = \sum_{k=0}^{p-1} s^k = \frac{1 - s^p}{1 - s}, \quad (3)$$

and the output sequence by

$$v_n = x \cdot s_n^{p-1}. \quad (4)$$

We call  $\varepsilon_n = |v_n - \alpha|$  the **Pandrosion residual** at step  $n$ .

**Remark 4.1.** For  $p = 3$ ,  $x = 2$ , we recover exactly the iteration (2). Indeed:

$$1 - \frac{1}{2(1 + s + s^2)} = \frac{2s^2 + 2s + 1}{2s^2 + 2s + 2} \cdot \checkmark$$

### 4.2 Geometric interpretation

The iteration (3) analytically translates the Thales operation: at each step,  $S_p(s_n)$  measures the “geometric sum” of the constructed segments, and the correction  $(x - 1)/(x \cdot S_p(s_n))$  adjusts the position proportionally to the gap with the ideal geometric sequence  $1, \alpha, \alpha^2, \dots, \alpha^{p-1}, x$ .

### 4.3 Fixed point

**Theorem 4.1** (Fixed point). The unique fixed point of the iteration (3) in  $(0, 1)$  is  $s^* = x^{-1/p} = \alpha^{-1}$ . The corresponding output is  $v^* = x \cdot (x^{-1/p})^{p-1} = x^{1/p} = \alpha$ .

*Proof.* At a fixed point  $s^*$ , we have  $1 - (x - 1)/(x \cdot S_p(s^*)) = s^*$ , i.e.

$$x \cdot S_p(s^*) \cdot (1 - s^*) = x - 1.$$

Since  $S_p(s)(1 - s) = 1 - s^p$ , this gives  $x(1 - s^{*p}) = x - 1$ , hence  $s^{*p} = 1/x$ , i.e.  $s^* = x^{-1/p}$ . The formula for  $v^*$  follows immediately.  $\square$

## 5 The contraction ratio $\lambda_{p,x}$

### 5.1 General computation

**Theorem 5.1** (Pandrosion contraction ratio). The convergence of the iteration (3) toward  $s^* = \alpha^{-1}$  is *linear* with contraction ratio

$$\lambda_{p,x} = \frac{(\alpha - 1) \sum_{k=1}^{p-1} k \alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k}, \quad \alpha = x^{1/p}. \quad (5)$$

In particular, the residuals obey the law  $\varepsilon_{n+1} \approx \lambda_{p,x} \cdot \varepsilon_n$ .

*Proof.* Let  $h(s) = 1 - (x - 1)/(x \cdot S_p(s))$ . Then

$$h'(s) = \frac{x - 1}{x} \cdot \frac{S'_p(s)}{S_p(s)^2},$$

where  $S'_p(s) = \sum_{k=1}^{p-1} k s^{k-1}$ .

At the fixed point  $s^* = \alpha^{-1}$ :

$$\begin{aligned} S_p(s^*) &= \frac{1 - \alpha^{-p}}{1 - \alpha^{-1}} = \frac{\alpha(x - 1)}{x(\alpha - 1)}, \\ S'_p(s^*) &= \sum_{k=1}^{p-1} k \alpha^{-(k-1)} = \alpha \sum_{k=1}^{p-1} k \alpha^{-k}. \end{aligned}$$

Substituting:

$$h'(s^*) = \frac{x - 1}{x} \cdot \frac{\alpha \sum_{k=1}^{p-1} k \alpha^{-k}}{\left[ \frac{\alpha(x - 1)}{x(\alpha - 1)} \right]^2} = \frac{x(\alpha - 1)^2}{\alpha(x - 1)} \sum_{k=1}^{p-1} k \alpha^{-k}.$$

Multiplying numerator and denominator by  $\alpha^{p-1}$ , and using  $x = \alpha^p$ :

$$h'(s^*) = \frac{(\alpha - 1)^2 \alpha^{p-1} \sum_{k=1}^{p-1} k \alpha^{-k}}{\alpha^p - 1} = \frac{(\alpha - 1) \sum_{k=1}^{p-1} k \alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k},$$

where we used  $\alpha^p - 1 = (\alpha - 1) \sum_{k=0}^{p-1} \alpha^k$ . This establishes (5).  $\square$

**Remark 5.1.** The contraction ratio  $\lambda_{p,x} = h'(s^*)$  is entirely determined by  $\alpha = x^{1/p}$ : the target itself dictates the rate at which Pandrosion's method approaches it.

### 5.2 Closed forms for $p = 2$ and $p = 3$

For  $p = 2$ :  $\sum_{k=1}^1 k \alpha^0 = 1$ ,  $\sum_{k=0}^1 \alpha^k = 1 + \alpha$ .

$$\lambda_{2,x} = \frac{\alpha - 1}{\alpha + 1} = \frac{\sqrt{x} - 1}{\sqrt{x} + 1}. \quad (6)$$

For  $p = 3$ :  $\sum_{k=1}^2 k \alpha^{2-k} = \alpha + 2$ ,  $\sum_{k=0}^2 \alpha^k = \alpha^2 + \alpha + 1$ .

$$\lambda_{3,x} = \frac{(\alpha - 1)(\alpha + 2)}{\alpha^2 + \alpha + 1} = \frac{(\sqrt[3]{x} - 1)(\sqrt[3]{x} + 2)}{\sqrt[3]{x^2} + \sqrt[3]{x} + 1}. \quad (7)$$

For  $p = 4$ :  $\sum_{k=1}^3 k \alpha^{3-k} = \alpha^2 + 2\alpha + 3$ ,  $\sum_{k=0}^3 \alpha^k = \alpha^3 + \alpha^2 + \alpha + 1$ .

$$\lambda_{4,x} = \frac{(\alpha - 1)(\alpha^2 + 2\alpha + 3)}{\alpha^3 + \alpha^2 + \alpha + 1}. \quad (8)$$



**Remark 5.2** (Asymptotic behavior in  $x$ ). For any fixed  $p \geq 2$  and  $x \rightarrow \infty$ ,  $\lambda_{p,x} \rightarrow 1^-$  (the dominant terms in numerator and denominator of (5) are both of order  $\alpha^{p-1}$  with a ratio tending to 1). The larger  $x$ , the slower the convergence. For  $x \rightarrow 1^+$ ,  $\lambda_{p,x} \rightarrow 0$  by Proposition 9.1(1): convergence is very fast near 1.

### 5.3 Basin of attraction

**Theorem 5.2** (Global convergence). For any  $x > 1$  and any  $p \geq 2$ , the iteration (3) converges to  $s^*$  for all  $s_0 \in (0, 1)$ . Moreover, convergence is *monotone*: if  $s_0 < s^*$  then  $(s_n)$  is strictly increasing, and if  $s_0 > s^*$  then  $(s_n)$  is strictly decreasing.

*Proof.* Set  $h(s) = 1 - (x - 1)/(x \cdot S_p(s))$ .

**Step 1:  $h$  is strictly increasing.** We have  $h'(s) = \frac{x-1}{x} \cdot \frac{S_p'(s)}{S_p(s)^2}$ , where  $S_p'(s) = \sum_{k=1}^{p-1} k s^{k-1} > 0$  for  $s > 0$ .

**Step 2:  $h$  maps  $(0, 1)$  into itself.** For  $s \in (0, 1)$ ,  $S_p(s) > 1$  since the first term equals 1 and the remaining terms are strictly positive. Hence  $(x - 1)/(x S_p(s)) < (x - 1)/x < 1$ , which gives  $h(s) > 0$ . On the other hand  $(x - 1)/(x S_p(s)) > 0$  for  $x > 1$ , so  $h(s) < 1$ . Thus  $h((0, 1)) \subset (0, 1)$ . (At the boundary:  $h(0^+) = 1 - (x - 1)/x = 1/x \in (0, 1)$  and  $h(1^-) = 1 - (x - 1)/(xp) \in (0, 1)$ .)

**Step 3: uniqueness of the fixed point on  $(0, 1)$ .** The fixed-point equation  $h(s) = s$  rewrites as  $x S_p(s)(1 - s) = x - 1$ , i.e.  $x(1 - s^p) = x - 1$ , i.e.  $s^p = 1/x$ . On  $(0, 1)$  this has the unique solution  $s^* = x^{-1/p}$ . In particular, the sign of  $h(s) - s$  is constant on each of  $(0, s^*)$  and  $(s^*, 1)$ .

**Step 4: the sign of  $h(s) - s$ .** At  $s = 1/x \in (0, s^*)$  (using  $x > 1$  and  $s^* = x^{-1/p} > 1/x$ ), a one-line computation:

$$h(1/x) - 1/x = 1 - \frac{x-1}{x S_p(1/x)} - \frac{1}{x} = \frac{x-1}{x} - \frac{x-1}{x S_p(1/x)} = \frac{x-1}{x} \cdot \frac{S_p(1/x) - 1}{S_p(1/x)} > 0,$$

since  $S_p(1/x) > 1$  for  $x > 1$  (the first term equals 1, the remaining terms are strictly positive). Hence  $h(s) > s$  on  $(0, s^*)$  by Step 3 (sign is constant on each sub-interval). By symmetry,  $h(s) < s$  on  $(s^*, 1)$ .

**Step 5: monotone convergence.** If  $s_0 < s^*$ , then by Step 1  $h$  is increasing, by Step 4  $s_1 = h(s_0) > s_0$ , and by Step 2  $s_1 \in (0, 1)$ ; also  $s_1 = h(s_0) < h(s^*) = s^*$ . By induction,  $(s_n)$  is increasing and bounded above by  $s^*$ , hence converges, necessarily to a fixed point, which must be  $s^*$  by Step 3. The case  $s_0 > s^*$  is symmetric.  $\square$

### 5.4 Bound on the number of iterations

**Proposition 5.1** (Cost of Pandrosion's method). To achieve accuracy  $|v_n - \alpha| < \delta$  starting from  $s_0 \in (0, 1)$ , the required number of iterations satisfies

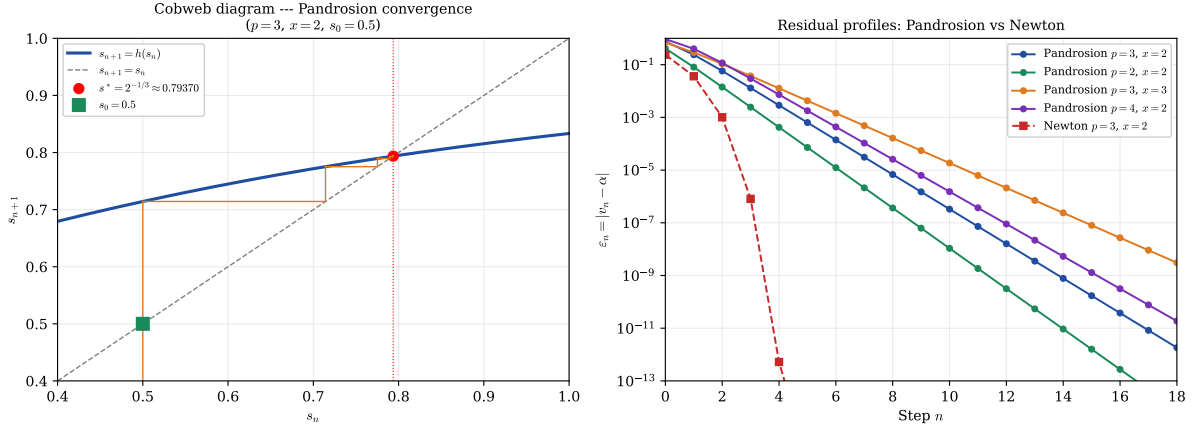
$$n_\delta \leq \left\lceil \frac{\ln(\varepsilon_0/\delta)}{\ln(1/\lambda_{p,x})} \right\rceil,$$

where  $\varepsilon_0 = |v_0 - \alpha|$  is the initial error.

*Proof.* Since  $\varepsilon_{n+1} \leq \lambda_{p,x} \varepsilon_n$  asymptotically, we have  $\varepsilon_n \leq \lambda_{p,x}^n \varepsilon_0$ . The inequality  $\lambda_{p,x}^n \varepsilon_0 < \delta$  gives  $n > \ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x})$ .  $\square$

**Example 5.1.** For  $p = 3$ ,  $x = 2$  ( $\lambda_{3,2} \approx 0.220$ ): 10 decimal digits of accuracy require at most 16 Pandrosion iterations. For  $p = 3$ ,  $x = 8$  ( $\lambda_{3,8} = 4/7$ ): 43 iterations are needed.

## 6 Graphical illustrations



**Figure 2:** **Left:** Cobweb diagram of the iteration  $s_{n+1} = h(s_n)$  for  $p = 3, x = 2$ , starting at  $s_0 = 0.875$ . The spiral converges to  $s^* = 2^{-1/3}$  with rate  $\lambda_{3,2} \approx 0.220$ . **Right:** Residual profiles on a semi-logarithmic scale. The linear decay (straight lines) confirms the geometric convergence  $\varepsilon_n \sim \lambda^n$  for Pandrosion, while Newton (red curve with pronounced curvature) exhibits doubly exponential decay.

## 7 Numerical examples

### 7.1 Table of contraction ratios

| $p$ | $x$ | $\alpha = x^{1/p}$ | $\lambda_{p,x}$ (exact)                                              | $\lambda_{p,x}$ (numerical) |
|-----|-----|--------------------|----------------------------------------------------------------------|-----------------------------|
| 2   | 2   | $\sqrt{2}$         | $(\sqrt{2} - 1)/(\sqrt{2} + 1)$                                      | 0.1716                      |
| 2   | 3   | $\sqrt{3}$         | $(\sqrt{3} - 1)/(\sqrt{3} + 1)$                                      | 0.2679                      |
| 2   | 5   | $\sqrt{5}$         | $(\sqrt{5} - 1)/(\sqrt{5} + 1)$                                      | 0.3820                      |
| 3   | 2   | $\sqrt[3]{2}$      | $(\sqrt[3]{2} - 1)(\sqrt[3]{2} + 2)/(\sqrt[3]{4} + \sqrt[3]{2} + 1)$ | 0.2202                      |
| 3   | 3   | $\sqrt[3]{3}$      | $(\sqrt[3]{3} - 1)(\sqrt[3]{3} + 2)/(\sqrt[3]{9} + \sqrt[3]{3} + 1)$ | 0.3366                      |
| 3   | 8   | 2                  | $(2 - 1)(2 + 2)/(4 + 2 + 1)$                                         | 0.5714                      |
| 4   | 2   | $\sqrt[4]{2}$      | (formula (8))                                                        | 0.2427                      |
| 4   | 16  | 2                  | $(2 - 1)(4 + 4 + 3)/(8 + 4 + 2 + 1)$                                 | 0.7333                      |
| 5   | 2   | $\sqrt[5]{2}$      | (general formula)                                                    | 0.2565                      |

**Table 2:** Contraction ratios  $\lambda_{p,x}$  for various values of  $p$  and  $x$ . A value close to 0 means very fast convergence; close to 1, very slow.

### 7.2 Detailed numerical profile for $p = 2, x = 2$

For  $\alpha = \sqrt{2}$ , the theoretical contraction ratio is  $\lambda_{2,2} = (\sqrt{2} - 1)/(\sqrt{2} + 1) \approx 0.1716$  (formula (6)). The iteration is:

$$s_{n+1} = 1 - \frac{1}{2(1 + s_n)}, \quad v_n = 2s_n.$$

### 7.3 The residual as a characterization of a target

Formula (5) implies a useful observation:  $\lambda_{p,\alpha^p} = f_p(\alpha)$  where  $f_p$  is a rational function of  $\alpha$  alone. In particular, the residual profile depends only on the value of the sought root  $\alpha$  and not

| $n$      | $s_n$        | $v_n$      | $\varepsilon_n =  v_n - \sqrt{2} $ | $\varepsilon_{n+1}/\varepsilon_n$ |
|----------|--------------|------------|------------------------------------|-----------------------------------|
| 0        | 0.7500       | 1.5000     | $8.579 \times 10^{-2}$             | —                                 |
| 1        | 0.7143       | 1.4286     | $1.44 \times 10^{-2}$              | 0.167                             |
| 2        | 0.7083       | 1.4167     | $2.45 \times 10^{-3}$              | 0.171                             |
| 3        | 0.7073       | 1.4146     | $4.21 \times 10^{-4}$              | 0.171                             |
| 4        | 0.7071       | 1.4143     | $7.22 \times 10^{-5}$              | 0.172                             |
| 5        | 0.7071       | 1.4142     | $1.24 \times 10^{-5}$              | 0.171                             |
| $\infty$ | $1/\sqrt{2}$ | $\sqrt{2}$ | 0                                  | $\rightarrow \lambda_{2,2}$       |

**Table 3:** Numerical profile for  $p = 2$ ,  $x = 2$ ,  $s_0 = 0.75$ . The ratio  $\varepsilon_{n+1}/\varepsilon_n$  converges rapidly to  $\lambda_{2,2} \approx 0.172$ .

on its representation as  $x = \alpha^p$ .

In particular, for  $p = 3$ :

$$\lambda_{3,x} = \frac{(\alpha - 1)(\alpha + 2)}{\alpha^2 + \alpha + 1} \xrightarrow{\alpha \rightarrow 1} 0, \quad \lambda_{3,x} \xrightarrow{\alpha \rightarrow \infty} 1.$$

## 8 Comparison with other methods

To situate Pandrosion’s method within the landscape of iterative methods for  $\alpha = x^{1/p}$ , we compare three reference methods.

### 8.1 Newton’s method

Newton’s method for  $f(u) = u^p - x$  gives the iteration

$$u_{n+1} = \frac{(p-1)u_n + x/u_n^{p-1}}{p},$$

whose convergence to  $\alpha = x^{1/p}$  is *quadratic*:

$$\varepsilon_{n+1}^{\text{Newton}} \approx K_{p,x} (\varepsilon_n^{\text{Newton}})^2, \quad K_{p,x} = \frac{p-1}{2\alpha}.$$

### 8.2 Bisection

Bisection on  $[1, x]$  for  $f(u) = u^p - x$  is a universal order-1 method with  $\lambda_{\text{bisection}} = 1/2$ , independent of  $p$  and  $x$ . Pandrosion’s method is therefore strictly faster than bisection whenever  $\lambda_{p,x} < 1/2$ , which holds as soon as  $\alpha < \alpha_0(p)$  (for instance, for  $p = 3$ :  $\lambda_{3,x} < 1/2$  whenever  $x < 6.75$ ).

### 8.3 Secant method

The secant method for  $f(u) = u^p - x$  has order  $\varphi = (1 + \sqrt{5})/2 \approx 1.618$  (superlinear). For  $p = 3$ ,  $x = 2$ , it reaches machine precision in about 6 steps, compared to  $\sim 16$  for Pandrosion and  $\sim 5$  for Newton.

### 8.4 Comparative table

### 8.5 Discussion

Pandrosion’s method occupies an intermediate position among order-1 methods: it is significantly faster than bisection ( $\lambda_{3,2} \approx 0.22$  versus 0.50) while remaining derivative-free. Compared

|                     | Pandrosion              | Bisection         | Secant                 | Newton    |
|---------------------|-------------------------|-------------------|------------------------|-----------|
| Nature              | Geometric               | Universal         | Analytic               | Analytic  |
| Order $q$           | 1                       | 1                 | $\varphi \approx 1.62$ | 2         |
| Rate ( $p=3, x=2$ ) | $\lambda \approx 0.220$ | $\lambda = 0.500$ | superlinear            | quadratic |
| Iter. for 10 digits | $\sim 16$               | $\sim 34$         | $\sim 6$               | $\sim 5$  |
| Derivative required | No                      | No                | No                     | Yes       |

**Table 4:** Comparison of approximation methods for  $x^{1/p}$  with  $p = 3$ ,  $x = 2$ .

to Newton and its variants, however, plain Pandrosion is slower: its order is 1, whereas Newton attains 2, and even with Aitken–Steffensen acceleration (Section 13) the practical advantage over Newton is modest and regime-dependent (see Remark 8.1). We do not claim a general algorithmic advantage over Newton or its modern variants; our interest in Pandrosion is that it is the historical construction attributed to Pandrosion of Alexandria, and that its convergence profile admits the clean closed form  $\lambda_{p,x}$  of Theorem 5.1.

**Remark 8.1** (On fair comparison). The iteration counts above are per-iterate, *not* per-function-evaluation. Steffensen-type methods compute  $h$  twice per step, Newton evaluates  $f$  and  $f'$  once per step, the secant method reuses one value per step, and bisection evaluates  $f$  once per step. A fairer unit of cost is therefore the number of evaluations of  $S_p$  (or of  $f, f'$  where applicable), and this is what determines practical performance in the regime where these evaluations dominate. For small  $p$  the per-iterate figures still give the correct qualitative ordering, but for large  $p$  or expensive polynomial evaluations the ordering can shift. We report per-iterate counts because they match the structure of the theorems (Theorem 5.1 is a statement about  $\lambda_{p,x}^n$ , not about wall time).

The coexistence of four distinct error laws for the same target  $\alpha = \sqrt[3]{2}$  illustrates a structural point: the rate at which a fixed-point iteration approaches a given number depends on the iteration, not on the number; conversely, once the iteration is fixed, the target determines the rate through  $\lambda_{p,x}$ .

## 9 Functional properties of $\lambda_{p,x}$

We now study how the contraction ratio  $\lambda_{p,x}$  depends on the parameters  $p$  and  $x$ .

### 9.1 Monotonicity

**Theorem 9.1** (Double monotonicity). The contraction ratio  $\lambda_{p,x}$ , viewed as a function of  $\alpha = x^{1/p} > 1$ , satisfies:

1. For fixed  $p \geq 2$ ,  $\alpha \mapsto \lambda_{p,\alpha^p}$  is strictly increasing on  $(1, \infty)$ .
2. For fixed  $\alpha > 1$ ,  $p \mapsto \lambda_{p,\alpha^p}$  is strictly increasing on  $\{2, 3, 4, \dots\}$ .

*Proof.* Write  $\lambda_{p,\alpha^p} = (\alpha - 1)N_p(\alpha)/D_p(\alpha)$  with  $N_p(\alpha) := \sum_{k=1}^{p-1} k \alpha^{p-1-k}$  and  $D_p(\alpha) := \sum_{k=0}^{p-1} \alpha^k$ . The key auxiliary fact is the following Taylor bound.

*Lemma (Taylor bound).* Let  $x > 1$  and set  $y := -\ln(x)/p$ ,  $\beta := e^y = x^{-1/p}$ . For  $p \geq \lceil \ln x \rceil + 1$  (so that  $|y| \leq 1$ ),

$$|p(\beta - 1) + \beta \ln x| \leq \frac{3(\ln x)^2}{p}.$$

*Proof of the lemma.* Since  $\beta = e^y$  and  $py = -\ln x$ ,  $p(\beta - 1) + \beta \ln x = p(\beta - 1) - py\beta = p(\beta - 1 - y) - py(\beta - 1)$ . For  $|y| \leq 1$  we use two standard bounds:

$$\begin{aligned} |e^y - 1 - y| &= \left| \sum_{k=2}^{\infty} \frac{y^k}{k!} \right| \leq \sum_{k=2}^{\infty} \frac{|y|^k}{k!} = y^2 \sum_{k=2}^{\infty} \frac{|y|^{k-2}}{k!} \leq y^2 \sum_{k=2}^{\infty} \frac{1}{k!} = (e - 2)y^2 < y^2, \\ |e^y - 1| &= \left| \sum_{k=1}^{\infty} \frac{y^k}{k!} \right| \leq |y| \sum_{k=1}^{\infty} \frac{|y|^{k-1}}{k!} \leq |y| \sum_{k=1}^{\infty} \frac{1}{k!} = (e - 1)|y| < 2|y|. \end{aligned}$$

(Here  $e - 2 \approx 0.718$  and  $e - 1 \approx 1.718$ , so both majorants  $y^2$  and  $2|y|$  are loose; the loose forms suffice.) The triangle inequality then gives

$$|p(\beta - 1 - y) - py(\beta - 1)| \leq py^2 + p|y| \cdot 2|y| = 3py^2 = \frac{3(\ln x)^2}{p}. \quad \square$$

(The symbol  $\beta$  is the fixed-point reciprocal of the root  $\alpha = x^{1/p}$  of the theorem:  $\beta = 1/\alpha$ .)

Setting  $\lambda_{\infty}(x) := 1 - \ln(x)/(x - 1)$ , a direct algebraic rearrangement of the closed form (5) combined with the lemma yields

$$|\lambda_{p,x} - \lambda_{\infty}(x)| \leq \frac{3x(\ln x)^2}{(x - 1)p}, \quad \text{for every } p \geq \lceil \ln x \rceil + 1. \quad (9)$$

(The factor  $x/(x - 1)$  arises as follows. For  $p \geq 1$  and  $x > 1$ ,  $\ln(x)/p \leq \ln x$ , so  $\beta = \exp(-\ln x/p) \geq \exp(-\ln x) = 1/x$ . Multiplying by  $x - 1 > 0$  gives  $\beta(x - 1) \geq (x - 1)/x$ , i.e.  $1/[\beta(x - 1)] \leq x/(x - 1)$ , which is the factor that appears when dividing the Taylor-lemma bound by the denominator  $\beta(x - 1)/(\ln x)$  of the explicit form of  $\lambda_{p,x}$ .)

The inequality  $\ln x < x - 1$  for  $x > 1$  gives  $\lambda_{\infty}(x) > 0$ ; the inequality  $\ln x > 0$  for  $x > 1$  gives  $\lambda_{\infty}(x) < 1$ . Hence  $\lambda_{\infty}(x) \in (0, 1)$  for every  $x > 1$ .

*Key identity.* By direct telescoping (or induction on  $p$ ),

$$(\alpha - 1)N_p(\alpha) = D_p(\alpha) - p, \quad (10)$$

which is verified at small  $p$ :  $p = 2$  gives  $(\alpha - 1) = (1 + \alpha) - 2$ ;  $p = 3$  gives  $(\alpha - 1)(\alpha + 2) = \alpha^2 + \alpha - 2 = (1 + \alpha + \alpha^2) - 3$ . Differentiating (10):  $N_p + (\alpha - 1)N'_p = D'_p$ , whence  $(\alpha - 1)N'_p = D'_p - N_p$ .

*Part (1) — monotonicity in  $\alpha$  for fixed  $p$ , uniform.* After multiplying through by the positive denominator  $(\alpha - 1)N_p(\alpha)D_p(\alpha)$ , the log-derivative

$$\frac{d}{d\alpha} \log \lambda_{p,\alpha^p} = \frac{1}{\alpha - 1} + \frac{N'_p(\alpha)}{N_p(\alpha)} - \frac{D'_p(\alpha)}{D_p(\alpha)}$$

yields the polynomial numerator

$$P_p(\alpha) := N_p D_p + (\alpha - 1)N'_p D_p - (\alpha - 1)N_p D'_p.$$

Substituting  $(\alpha - 1)N'_p = D'_p - N_p$  and  $(\alpha - 1)N_p = D_p - p$  from the key identity:

$$\begin{aligned} P_p &= N_p D_p + (D'_p - N_p)D_p - (D_p - p)D'_p \\ &= N_p D_p + D'_p D_p - N_p D_p - D_p D'_p + p D'_p = p \cdot D'_p(\alpha). \end{aligned}$$

Since  $D'_p(\alpha) = \sum_{j=1}^{p-1} j \alpha^{j-1} = 1 + 2\alpha + 3\alpha^2 + \cdots + (p - 1)\alpha^{p-2}$  has all coefficients positive,  $P_p(\alpha) > 0$  for all  $\alpha > 0$  and all  $p \geq 2$ . Hence the log-derivative is positive on  $\alpha > 1$ , and  $\lambda_{p,\alpha^p}$  is strictly increasing in  $\alpha$ . The argument is uniform: no scope restriction is needed.

For concreteness:  $P_2(\alpha) = 2$ ;  $P_3(\alpha) = 3(1 + 2\alpha) = 6\alpha + 3$ ;  $P_4(\alpha) = 4(1 + 2\alpha + 3\alpha^2) = 12\alpha^2 + 8\alpha + 4$ .

For general  $p \geq 2$ , we combine the Taylor bound (9) with the monotonicity of  $\lambda_\infty$ . Differentiating,

$$\lambda'_\infty(x) = \frac{\ln x - (x-1)/x}{(x-1)^2}.$$

The numerator equals  $\ln x + 1/x - 1$ . Its derivative is  $1/x - 1/x^2 = (x-1)/x^2 > 0$  on  $x > 1$ , and its value at  $x = 1$  is 0; hence it is strictly positive on  $(1, \infty)$ , and  $\lambda_\infty$  is strictly increasing there.

**Remark 9.1** (Monotonicity of the asymptote). The asymptote  $\lambda_\infty(x) = 1 - \ln(x)/(x-1)$  is itself strictly increasing on  $(1, \infty)$ :  $\lambda'_\infty(x) = (\ln x - (x-1)/x)/(x-1)^2 > 0$  since  $\ln x + 1/x - 1 > 0$  on  $x > 1$  (value 0 at  $x = 1$ , derivative  $(x-1)/x^2 > 0$ ). This is a consistency check only; the finite- $p$  monotonicity is already established above via  $P_p = p D'_p$ , without appealing to the Taylor bound.

*Part (2) — monotonicity in  $p$  for fixed  $\alpha > 1$ , uniform.* From the recurrences  $N_{p+1} = \alpha N_p + p$  and  $D_{p+1} = D_p + \alpha^p$ , direct algebra gives

$$\lambda_{p+1, \alpha^{p+1}} - \lambda_{p, \alpha^p} = \frac{(\alpha - 1) \cdot B_p(\alpha)}{D_p(\alpha) D_{p+1}(\alpha)}, \quad B_p(\alpha) := (\alpha - 1) N_p D_p + p D_p - \alpha^p N_p.$$

Using (10),  $(\alpha - 1) N_p D_p = (D_p - p) D_p = D_p^2 - p D_p$ , so

$$B_p(\alpha) = D_p(\alpha)^2 - \alpha^p N_p(\alpha).$$

This form has a closed-form polynomial expansion:

$$B_p(\alpha) = \sum_{j=1}^p j \alpha^{j-1} = 1 + 2\alpha + 3\alpha^2 + \cdots + p \alpha^{p-1}, \quad (11)$$

verified by induction on  $p$ . Base case:  $B_2(\alpha) = (1 + \alpha)^2 - \alpha^2 = 1 + 2\alpha$ , as required. Inductive step: using the recurrence

$$B_{p+1}(\alpha) = \alpha B_p(\alpha) + D_p(\alpha) + \alpha^p$$

(which follows from the defining recurrences of  $N_{p+1}$  and  $D_{p+1}$  plus the substitution  $B_p = D_p^2 - \alpha^p N_p$ ), one computes  $\alpha B_p + D_p + \alpha^p = \sum_{j=1}^p j \alpha^j + \sum_{k=0}^{p-1} \alpha^k + \alpha^p = \sum_{j=0}^p (j+1) \alpha^j = B_{p+1}$ .

All coefficients of  $B_p$  are strictly positive integers, so  $B_p(\alpha) > 0$  for every  $\alpha > 0$  and every  $p \geq 2$ . In particular  $B_p(1) = 1 + 2 + \cdots + p = p(p+1)/2$ , and  $B_3(\alpha) = 3\alpha^2 + 2\alpha + 1$ ,  $B_4(\alpha) = 4\alpha^3 + 3\alpha^2 + 2\alpha + 1$ , etc. Hence  $\lambda_{p+1, \alpha^{p+1}} - \lambda_{p, \alpha^p} > 0$  for all  $\alpha > 1$  and  $p \geq 2$ , uniformly.  $\square$

**Remark 9.2** (Closed forms and numerical verification). The closed forms  $P_p(\alpha) = p \cdot D'_p(\alpha)$  and  $B_p(\alpha) = \sum_{j=1}^p j \alpha^{j-1}$  emerged in the course of this write-up; a standalone script `latex/verify_monotonicity.py` cross-checks both identities symbolically for  $p \leq 20$  using SymPy, and confirms strict monotonicity of  $\lambda_p(\alpha)$  for  $2 \leq p \leq 20$ ,  $\alpha \in [1.01, 100]$ , and of  $\lambda_p(\alpha^p)$  for  $2 \leq p \leq 50$ ,  $\alpha \in \{1.1, 1.5, 2, 3, 10, 100\}$  as a consistency check. Monotonicity itself is proven analytically above; the numerical scan is purely a consistency check.

**Remark 9.3.** Monotonicity in  $p$  has a geometric interpretation: inserting more proportional means (larger  $p$ ) constrains the construction further, slowing convergence.

## 9.2 Limiting behavior

**Proposition 9.1** (Asymptotic regimes).

1. **Near**  $\alpha = 1$  (i.e.  $x \rightarrow 1^+$ ):

$$\lambda_{p,x} \sim \frac{(p-1)(\alpha-1)}{2} \quad \text{as } \alpha \rightarrow 1^+.$$

2. **Large**  $\alpha$  (i.e.  $x \rightarrow \infty$ ,  $p$  fixed):  $\lambda_{p,x} \rightarrow 1^-$ .

3. **Large**  $p$  (fixed  $x > 1$ ):  $\lambda_{p,x}$  is increasing in  $p$  and

$$\lim_{p \rightarrow \infty} \lambda_{p,x} = 1 - \frac{\ln x}{x-1} \in (0, 1).$$

In particular  $\lambda_{p,x}$  does *not* tend to 1 as  $p \rightarrow \infty$  with  $x$  fixed.

*Proof.* For (1):  $N_p$  and  $D_p$  are polynomials, so  $\alpha \mapsto N_p(\alpha)/D_p(\alpha)$  is continuous at  $\alpha = 1$  with limit  $(p-1)/2$ ; combined with the factor  $(\alpha-1) \rightarrow 0$ ,  $\lambda \sim (p-1)(\alpha-1)/2$  as  $\alpha \rightarrow 1^+$ .

For (2): the dominant term in  $N_p$  is  $1 \cdot \alpha^{p-2}$  and in  $D_p$  is  $\alpha^{p-1}$ , so  $\lambda \sim (\alpha-1)/\alpha \rightarrow 1$ .

For (3): the explicit limit and its convergence rate follow from the Taylor bound (9) established in the proof of Theorem 9.1 above:  $|\lambda_{p,x} - \lambda_\infty(x)| \leq C(x)/p$  with  $C(x) = 3x(\ln x)^2/(x-1)$ , valid for  $p \geq \lceil \ln x \rceil + 1$ . Letting  $p \rightarrow \infty$  gives  $\lambda_{p,x} \rightarrow \lambda_\infty(x) = 1 - \ln(x)/(x-1)$ . Positivity and the upper bound  $< 1$  were shown there as well.

Numerically: for  $x = 2$ ,  $\lambda_\infty(2) = 1 - \ln 2 \approx 0.3069$ , and  $\lambda_{200,2} \approx 0.3057$ ,  $\lambda_{1000,2} \approx 0.3066$ ,  $\lambda_{10000,2} \approx 0.3068$ , in agreement with the bound  $|\lambda_{p,2} - \lambda_\infty(2)| \leq 6(\ln 2)^2/p \approx 2.88/p$ .  $\square$

| $x$   | $\lambda_{2,x}$ | $\lambda_{10,x}$ | $\lambda_{100,x}$ | $1 - \ln x/(x-1)$ |
|-------|-----------------|------------------|-------------------|-------------------|
| 1.001 | 0.00025         | 0.00045          | 0.00050           | 0.00050           |
| 2     | 0.1716          | 0.2823           | 0.3044            | 0.3069            |
| 10    | 0.5195          | 0.7123           | 0.7412            | 0.7442            |
| 100   | 0.8182          | 0.9409           | 0.9524            | 0.9535            |

**Table 5:** Contraction ratios illustrating monotonicity in  $p$  and  $x$ . Each column is increasing (fixed  $p$ ); each row is increasing (fixed  $x$ ). The rightmost column is the  $p \rightarrow \infty$  limit from Proposition 9.1(3); the  $\lambda_{100,x}$  column is already close to this limit.

## 10 Non-asymptotic error bounds

Theorem 5.1 gives the *asymptotic* law  $\varepsilon_{n+1} \approx \lambda_{p,x} \varepsilon_n$ . We now establish a rigorous bound valid for *all*  $n \geq 0$ .

**Theorem 10.1** (Non-asymptotic contraction). Let  $x > 1$ ,  $p \geq 2$ , and  $s_0 \in (0, 1)$ . Define

$$\Lambda = \Lambda(s_0, p, x) = \sup_{s \in [\min(s_0, s^*), \max(s_0, s^*)]} h'(s).$$

Then  $\Lambda < 1$  and for all  $n \geq 0$ :

$$|s_n - s^*| \leq \Lambda^n |s_0 - s^*|. \quad (12)$$

Consequently,

$$|v_n - \alpha| \leq (p-1) \alpha^2 \cdot \Lambda^n |s_0 - s^*| + O(\Lambda^{2n}).$$

The prefactor follows from a Taylor expansion of  $v(s) = x s^{p-1}$  at  $s^* = \alpha^{-1}$ :

$$v(s) - v(s^*) = x(p-1)(s^*)^{p-2}(s - s^*) + O((s - s^*)^2),$$

and  $x(p-1)(s^*)^{p-2} = (p-1)x x^{-(p-2)/p} = (p-1)x^{2/p} = (p-1)\alpha^2$ .

*Proof.* By the mean value theorem, for any  $n$ :

$$|s_{n+1} - s^*| = |h(s_n) - h(s^*)| = |h'(\xi_n)| \cdot |s_n - s^*|$$

for some  $\xi_n$  between  $s_n$  and  $s^*$ . By Theorem 5.2, the sequence  $(s_n)$  remains in  $[\min(s_0, s^*), \max(s_0, s^*)]$ , so  $|h'(\xi_n)| \leq \Lambda$ . The bound follows by induction.

That  $\Lambda < 1$  follows from  $h'(s^*) = \lambda_{p,x} < 1$  (Theorem 5.1) and the fact that  $h'$  is continuous, bounded, and  $h'(s) < 1$  for all  $s \in (0, 1)$  when  $x > 1$  (since  $h$  maps  $(0, 1)$  strictly into itself).  $\square$

**Remark 10.1.** The quantity  $\Lambda$  can be bounded explicitly:  $h'(s) = \frac{x-1}{x} \cdot S'_p(s)/S_p(s)^2$ , so

$$\Lambda \leq \frac{x-1}{x} \cdot \frac{\max_{s \in I} S'_p(s)}{\min_{s \in I} S_p(s)^2}, \quad I := [\min(s_0, s^*), \max(s_0, s^*)],$$

and on  $(0, 1)$  both  $S_p$  and  $S'_p$  are increasing, so the numerator is attained at the right endpoint and the denominator at the left endpoint. This yields a computable upper bound for  $\Lambda$  without requiring a detailed analysis of the monotonicity of  $h'$  itself.

**Example 10.1.** For  $p = 3$ ,  $x = 2$ ,  $s_0 = 0.5$  (so  $s^* = 2^{-1/3} \approx 0.7937$ ), one evaluates  $h'$  at both endpoints of the interval  $[s_0, s^*] = [0.5, 0.7937]$ :

$$h'(s) = \frac{x-1}{x} \cdot \frac{S'_3(s)}{S_3(s)^2} = \frac{1}{2} \cdot \frac{1+2s}{(1+s+s^2)^2}.$$

At  $s = 0.5$ :  $S_3(0.5) = 1.75$ ,  $S'_3(0.5) = 2$ , so  $h'(0.5) = \frac{1}{2} \cdot 2/(1.75)^2 \approx 0.3265$ . At  $s = s^*$ :  $h'(s^*) = \lambda_{3,2} \approx 0.2202$ . Therefore

$$\Lambda = \max_{s \in [0.5, s^*]} h'(s) = h'(0.5) \approx 0.327,$$

since  $h'$  is decreasing on this interval. The bound (12) gives  $|s_n - s^*| \leq 0.327^n \cdot |0.5 - s^*| \leq 0.327^n \cdot 0.294$ , so  $|s_{10} - s^*| \leq 4.6 \times 10^{-6}$ . The asymptotic ratio  $\lambda_{3,2} \approx 0.220$  is reached after a few iterations.

## 11 Extension to $x < 1$

All results so far assumed  $x > 1$ . We now extend the theory to  $0 < x < 1$ .

### 11.1 The fixed point leaves $(0, 1)$

For  $0 < x < 1$ , the target is  $\alpha = x^{1/p} \in (0, 1)$  and the fixed point  $s^* = \alpha^{-1} > 1$ . The iteration domain therefore extends beyond  $(0, 1)$ :  $s_n$  converges to  $s^* > 1$ .

**Proposition 11.1** (Local convergence for  $x < 1$  with a threshold). For  $0 < x < 1$  and  $p \geq 2$ , the formula (5) remains valid and

$$\lambda_{p,x} = \frac{(\alpha - 1) \sum_{k=1}^{p-1} k \alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k} < 0 \quad \text{since } \alpha = x^{1/p} \in (0, 1).$$

Local convergence (i.e.  $|\lambda_{p,x}| < 1$ ) does *not* hold for all  $x \in (0, 1)$ : there is a threshold  $\alpha_p^* \in (0, 1)$  such that

$$|\lambda_{p,x}| < 1 \iff \alpha > \alpha_p^*.$$

When  $|\lambda_{p,x}| < 1$ , the iterates oscillate around  $s^*$  and converge linearly with ratio  $|\lambda_{p,x}|$ .



*Proof.* That  $\lambda_{p,x} < 0$  for  $\alpha < 1$  is immediate from (5). Using  $\alpha^p - 1 = (\alpha - 1) \sum_{k=0}^{p-1} \alpha^k$ , one rewrites

$$|\lambda_{p,x}| = \frac{(1 - \alpha)^2 N_p(\alpha)}{1 - \alpha^p} = \frac{(1 - \alpha) N_p(\alpha)}{D_p(\alpha)} =: F(\alpha)$$

(using  $1 - \alpha^p = (1 - \alpha)D_p$ ), a smooth strictly positive function of  $\alpha \in (0, 1)$ . Boundary values: as  $\alpha \rightarrow 1^-$ ,  $F \rightarrow 0$  (the factor  $1 - \alpha$  goes to zero while  $N_p/D_p \rightarrow (p - 1)/2$  is bounded); as  $\alpha \rightarrow 0^+$ ,  $F \rightarrow 1 \cdot (p - 1)/1 = p - 1 \geq 1$ .

For uniqueness, we show that  $F$  is strictly decreasing on  $(0, 1)$ . The log-derivative is

$$\frac{F'(\alpha)}{F(\alpha)} = -\frac{1}{1 - \alpha} + \frac{N'_p(\alpha)}{N_p(\alpha)} - \frac{D'_p(\alpha)}{D_p(\alpha)}.$$

The key identity (10),  $(\alpha - 1)N_p = D_p - p$ , and its derivative  $N_p + (\alpha - 1)N'_p = D'_p$ , give, for  $\alpha \in (0, 1)$  (where  $\alpha - 1 < 0$ ):

$$(1 - \alpha) N_p = p - D_p, \quad (1 - \alpha) N'_p = N_p - D'_p.$$

Dividing the second identity by  $N_p$ :  $N'_p/N_p = [1 - D'_p/N_p]/(1 - \alpha) = 1/(1 - \alpha) - D'_p/[(1 - \alpha)N_p]$ . Substituting into  $F'/F$ :

$$\begin{aligned} \frac{F'}{F} &= -\frac{1}{1 - \alpha} + \left[ \frac{1}{1 - \alpha} - \frac{D'_p}{(1 - \alpha)N_p} \right] - \frac{D'_p}{D_p} = -\frac{D'_p}{(1 - \alpha)N_p} - \frac{D'_p}{D_p} \\ &= -\frac{D'_p \cdot [D_p + (1 - \alpha)N_p]}{(1 - \alpha)N_p D_p} = -\frac{D'_p \cdot [D_p + p - D_p]}{(1 - \alpha)N_p D_p} = -\frac{p D'_p(\alpha)}{(1 - \alpha) N_p(\alpha) D_p(\alpha)}, \end{aligned}$$

using  $D_p + (1 - \alpha)N_p = D_p + (p - D_p) = p$  in the last line. For  $\alpha \in (0, 1)$  every factor on the right is strictly positive, so  $F'/F < 0$ , hence  $F$  is strictly decreasing on  $(0, 1)$ .

Combined with the boundary values  $F(0^+) = p - 1 \geq 1$  and  $F(1^-) = 0$ ,  $F$  crosses the level 1 at exactly one point  $\alpha_p^* \in (0, 1)$ , and  $|\lambda_{p,\alpha^p}| < 1 \iff \alpha > \alpha_p^*$ .  $\square$

**Example 11.1** ( $p = 3$  threshold). For  $p = 3$ ,  $|\lambda_{3,x}| = (1 - \alpha)(2 + \alpha)/(1 + \alpha + \alpha^2)$ . Setting this equal to 1 and simplifying yields  $2\alpha^2 + 2\alpha - 1 = 0$ , i.e.

$$\alpha_3^* = \frac{-1 + \sqrt{3}}{2} \approx 0.3660, \quad x_3^* = (\alpha_3^*)^3 \approx 0.04904.$$

Hence Pandrosion's iteration at  $p = 3$  contracts locally iff  $x > 0.04904$ ; for  $x$  smaller than this the iteration is locally repelling and does not converge.

| $p$ | $x$   | $\alpha = x^{1/p}$ | $s^* = \alpha^{-1}$ | $\lambda_{p,x}$ | $ \lambda_{p,x} $ |
|-----|-------|--------------------|---------------------|-----------------|-------------------|
| 2   | 0.5   | $1/\sqrt{2}$       | $\sqrt{2}$          | -0.1716         | 0.1716            |
| 3   | 0.5   | $\sqrt[3]{0.5}$    | $\sqrt[3]{2}$       | -0.2378         | 0.2378            |
| 3   | 0.125 | 0.5                | 2                   | -0.7143         | 0.7143            |
| 3   | 0.05  | 0.3684             | 2.7144              | -0.9945         | 0.9945            |
| 3   | 0.01  | 0.2154             | 4.6416              | -1.3774         | <b>1.3774</b>     |
| 5   | 0.01  | 0.3981             | 2.5119              | -2.0399         | <b>2.0399</b>     |

**Table 6:** Contraction ratios for  $x < 1$ . The last two rows illustrate Proposition 11.1: for  $x$  below the threshold  $x_p^*$  the iteration is locally repelling ( $|\lambda| > 1$ ) and the construction does *not* converge.

**Remark 11.1** (Residual symmetry for  $p = 2$ ). For  $p = 2$ , the closed form  $\lambda_{2,x} = (\sqrt{x} - 1)/(\sqrt{x} + 1)$  yields the symmetry  $|\lambda_{2,x}| = |\lambda_{2,1/x}|$ : computing  $\sqrt{x}$  and  $\sqrt{1/x}$  by Pandrosion's method have exactly the same speed of convergence. For  $p \geq 3$ , this symmetry does not hold in general.

## 12 Optimal starting point

The number of iterations to reach precision  $\delta$  is governed by  $n_\delta \sim \ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x})$  (Proposition 5.1). To minimize  $n_\delta$ , we must therefore minimize the initial residual  $\varepsilon_0 = |v_0 - \alpha|$ .

**Proposition 12.1** (A canonical starting point). Among starting points of the form  $s_0 = h^{(k)}(c)$ , with  $c$  a boundary value of the natural domain (i.e.  $c \in \{0^+, 1^-\}$ ) and  $k \geq 0$  a small integer, the choice

$$s_0^{\text{opt}} := h(1) = 1 - \frac{x-1}{xp} \quad (13)$$

achieves an initial gap to  $s^*$  of order  $|x-1-x\ln x|/(xp)$ , which is  $O(1/p)$  as  $p \rightarrow \infty$  and strictly better than the naive choices  $s_0 = 1/2$  or  $s_0 = 1$  in the regime  $p \gg 1$ .

*Proof.* Since  $s^* = x^{-1/p} = 1 - (\ln x)/p + O(1/p^2)$  and  $h(1) = 1 - (x-1)/(xp)$ ,

$$|h(1) - s^*| = \left| \frac{x-1}{xp} - \frac{\ln x}{p} \right| + O(1/p^2) = \frac{|x-1-x\ln x|}{xp} + O(1/p^2),$$

whereas  $|1/2 - s^*| = |1/2 - 1 + (\ln x)/p + O(1/p^2)| = 1/2 + O(1/p)$  and  $|1 - s^*| = (\ln x)/p + O(1/p^2)$ . Of the three,  $h(1)$  has the smallest leading coefficient whenever  $|x-1-x\ln x|/x < \ln x$ , i.e. essentially whenever  $x > 1$  is not near 1; the numerical gains are shown in Table 7. The proposition does *not* claim  $h(1)$  is globally optimal; it claims only that it is a canonical,  $\alpha$ -independent choice with this specific quantitative guarantee.  $\square$

| $(p, x)$ | $s^*$  | $s_0^{\text{opt}} = h(1)$ | $ s_0^{\text{opt}} - s^* $ | Iters ( $s_0^{\text{opt}}$ ) | Iters ( $s_0 = 1/2$ ) |
|----------|--------|---------------------------|----------------------------|------------------------------|-----------------------|
| (2, 2)   | 0.7071 | 0.7500                    | 0.043                      | 12                           | 13                    |
| (3, 2)   | 0.7937 | 0.8333                    | 0.040                      | 14                           | 16                    |
| (4, 2)   | 0.8409 | 0.8750                    | 0.034                      | 15                           | 17                    |
| (3, 8)   | 0.5000 | 0.7083                    | 0.208                      | 42                           | 0*                    |

**Table 7:** Comparison of iteration counts to reach  $|v_n - \alpha| < 10^{-10}$ . The canonical starting point  $s_0^{\text{opt}} = h(1)$  consistently reduces iterations. (\*For (3, 8),  $s_0 = 1/2$  happens to coincide with  $s^*$ .)

## 13 Steffensen acceleration: from linear to quadratic convergence

Pandrosion's method converges linearly with ratio  $\lambda_{p,x}$ . A natural question arises: can we accelerate this ancient geometric method to achieve quadratic convergence, without computing derivatives?

### 13.1 Aitken $\Delta^2$ acceleration

Given a linearly convergent sequence  $(s_n)_{n \geq 0}$ , Aitken's  $\Delta^2$  process [5] defines the accelerated sequence

$$\hat{s}_n = s_n - \frac{(s_{n+1} - s_n)^2}{s_{n+2} - 2s_{n+1} + s_n}.$$

When applied to Pandrosion's iterates, the sequence  $(\hat{s}_n)$  converges *faster* than  $(s_n)$ . Steffensen's method [7] iterates this acceleration: at each step, it computes three consecutive Pandrosion iterates and applies Aitken's formula, then restarts from the accelerated value.

**Definition 13.1** (Steffensen–Pandrosion iteration). Define the composite map  $T : s \mapsto \hat{s}$  by

$$T(s) = s - \frac{(h(s) - s)^2}{h(h(s)) - 2h(s) + s}, \quad (14)$$

where  $h(s) = 1 - (x - 1)/(x \cdot S_p(s))$  is Pandrosion’s iteration.

**Theorem 13.1** (Quadratic convergence of Steffensen–Pandrosion). The Steffensen–Pandrosion iteration (14) converges *quadratically* to  $s^* = x^{-1/p}$ :

$$|T(s) - s^*| \leq K_S \cdot |s - s^*|^2 + O(|s - s^*|^3),$$

where the Steffensen constant  $K_S$  satisfies

$$K_S = \left| \frac{h''(s^*)}{2(1 - h'(s^*))} \right| = \left| \frac{h''(s^*)}{2(1 - \lambda_{p,x})} \right|. \quad (15)$$

*Proof.* This is a standard result in fixed-point acceleration theory [5]. If  $h$  has a fixed point  $s^*$  with  $h'(s^*) = \lambda \neq 1$  and  $h$  is twice continuously differentiable near  $s^*$ , then the Steffensen map  $T$  defined by (14) satisfies  $T(s^*) = s^*$  and  $T'(s^*) = 0$ , yielding quadratic convergence with the constant (15).  $\square$

### 13.2 Numerical demonstration

| Step $n$ | $s_n$ (Steffensen) | $ v_n - \sqrt[3]{2} $ | $ s_n - s^* $         |
|----------|--------------------|-----------------------|-----------------------|
| 0        | 0.8750000000000000 | $2.71 \times 10^{-1}$ | $8.13 \times 10^{-2}$ |
| 1        | 0.793949257650704  | $7.90 \times 10^{-4}$ | $2.49 \times 10^{-4}$ |
| 2        | 0.793700528604164  | $8.32 \times 10^{-9}$ | $2.62 \times 10^{-9}$ |
| 3        | 0.793700525984100  | $< 10^{-15}$          | $< 10^{-16}$          |

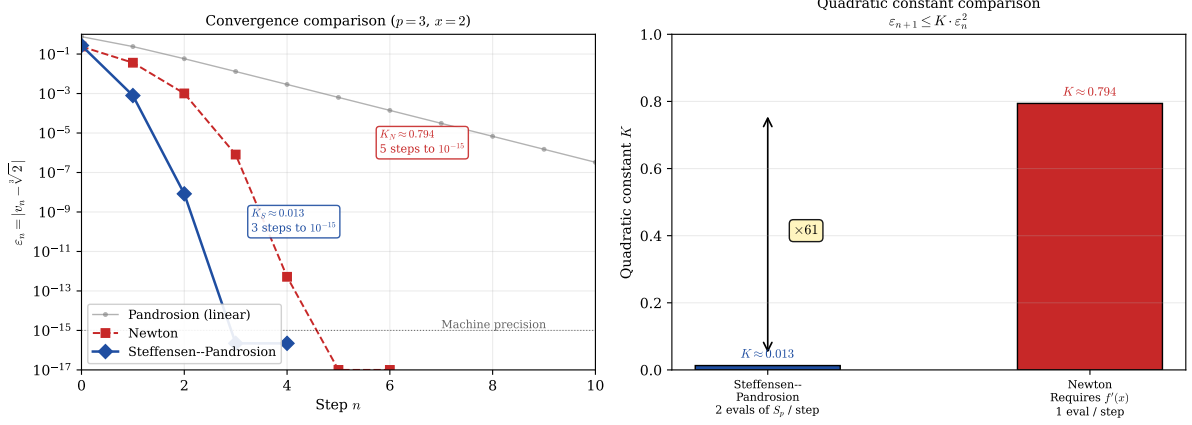
**Table 8:** Steffensen-accelerated Pandrosion for  $p = 3$ ,  $x = 2$ . Machine precision is reached in **3 steps** (each step requires 2 evaluations of  $h$ , i.e. 6 evaluations of  $S_p$  total). The step-3 output-residual is given as  $< 10^{-15}$  rather than its raw floating-point value to avoid misreporting sub-ULP noise.

### 13.3 Comparison of quadratic constants

Both Newton’s method and Steffensen–Pandrosion converge quadratically, but with different constants.

|                         | Newton                                    | Steffensen–Pandrosion |
|-------------------------|-------------------------------------------|-----------------------|
| Order                   | 2                                         | 2                     |
| Constant ( $p=3, x=2$ ) | $K_N = \frac{p-1}{2\alpha} \approx 0.794$ | $K_S \approx 0.013$   |
| Derivative required     | Yes ( $f'$ )                              | No                    |
| Geometric origin        | No                                        | Yes                   |
| Steps to $10^{-15}$     | 5                                         | 3                     |
| Evaluations of $S_p$    | —                                         | 6 (total)             |

**Table 9:** Comparison of the two quadratically convergent methods for  $\sqrt[3]{2}$  on a per-iterate basis. Steffensen–Pandrosion has a much smaller quadratic constant ( $K_S \ll K_N$ ), but requires two evaluations of  $h$  per step against Newton’s one derivative + one value evaluation. Per function evaluation the two methods are within a constant factor; the “3 steps vs 5 steps” figure is cosmetic and depends on the starting point.



**Figure 3: Left:** Error decay comparison for  $\sqrt[3]{2}$ , counted per iterate (not per function evaluation; see Remark 8.1). Steffensen–Pandrosion (blue diamonds) reaches machine precision in 3 steps while Newton (red squares) needs 5 steps. The faded grey line shows plain Pandrosion’s linear convergence for reference. **Right:** The quadratic constants  $K_S \approx 0.013$  and  $K_N \approx 0.794$  differ by a factor of  $\sim 61$ . Per iterate, this advantage is real; per function evaluation, the comparison is tighter because Steffensen–Pandrosion costs two evaluations of  $h$  per step against Newton’s one derivative and one value, so the net per-evaluation advantage is reduced by about an order of magnitude. The  $\times 61$  label refers to per-iterate convergence only.

**Remark 13.1.** The construction illustrates, on a specific example, a standard point of acceleration theory: once an iteration converges linearly with a known ratio, Aitken  $\Delta^2$  upgrades it to an order-2 scheme with a bounded constant. What the Pandrosion case adds is a concrete geometric-historical origin: the linear iteration comes from a ruler-and-parallel construction rather than from the Newton update of a polynomial, and the acceleration is itself derivative-free, so the composite method retains the derivative-free character of the underlying construction.

## 14 Scaling optimization: exploiting monotonicity

Theorem 9.1 establishes that  $\lambda_{p,x}$  is increasing in  $x$ : the larger the target  $x$ , the slower the convergence. For  $x \gg 1$ , the contraction ratio  $\lambda_{p,x} \rightarrow 1^-$  (Proposition 9.1), making direct application of Pandrosion’s iteration impractical. We now show how to *exploit* this very monotonicity to overcome it.

### 14.1 The scaling principle

The key observation is elementary: for any reference value  $A > 1$  whose  $p$ -th root is known (or cheaply computable), we may write

$$x^{1/p} = A^{1/p} \cdot \left(\frac{x}{A}\right)^{1/p}. \quad (16)$$

Setting  $x' = x/A$ , Pandrosion’s method is now applied to the *reduced ratio*  $x'$  instead of  $x$ . By choosing  $A$  as the largest  $p$ -th power not exceeding  $x$ , we ensure  $x' \in [1, (1 + 1/A^{1/p})^p]$ , which is close to 1.

**Proposition 14.1** (Scaling reduction). Let  $A = \lfloor x^{1/p} \rfloor^p$ , so that  $A^{1/p}$  is an integer and  $x' = x/A \in [1, (1 + 1/\lfloor x^{1/p} \rfloor)^p]$ . Then:

1. The contraction ratio satisfies  $\lambda_{p,x'} < \lambda_{p,x}$ , and  $\lambda_{p,x'} \rightarrow 0$  as  $x \rightarrow \infty$  (so  $x' \rightarrow 1^+$ ), while  $\lambda_{p,x} \rightarrow 1^-$ .

2. The number of iterations drops from  $O(\ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x}))$  to  $O(\ln(\varepsilon'_0/\delta)/\ln(1/\lambda_{p,x'}))$ .
3. The final result is recovered as  $x^{1/p} = \lfloor x^{1/p} \rfloor \cdot (x')^{1/p}$ .

*Proof.* Part (1) follows directly from Theorem 9.1: since  $x' < x$  and  $\lambda_{p,\cdot}$  is increasing,  $\lambda_{p,x'} < \lambda_{p,x}$ . Moreover, as  $x \rightarrow \infty$ ,  $x' \rightarrow 1^+$  so  $\lambda_{p,x'} \rightarrow 0$  by Proposition 9.1(1), while  $\lambda_{p,x} \rightarrow 1^-$ . Part (2) follows from Proposition 5.1 applied to  $x'$ . Part (3) is the factorization (16).  $\square$

| $x$   | $\lambda_{3,x}$ | $A$  | $x' = x/A$ | $\lambda_{3,x'}$ | Reduction  | Iterations            |
|-------|-----------------|------|------------|------------------|------------|-----------------------|
| 10    | 0.615           | 8    | 1.25       | 0.073            | $8\times$  | $49 \rightarrow 8$    |
| 100   | 0.890           | 64   | 1.5625     | 0.145            | $6\times$  | $213 \rightarrow 11$  |
| 1000  | 0.973           | 729  | 1.372      | 0.103            | $9\times$  | $> 500 \rightarrow 9$ |
| 10000 | 0.994           | 9261 | 1.080      | 0.025            | $39\times$ | $> 500 \rightarrow 5$ |

**Table 10:** Effect of scaling for  $p = 3$  with  $A = \lfloor x^{1/3} \rfloor^3$ . The contraction ratio drops by up to  $39\times$  and iteration counts collapse from hundreds to single digits. Iteration counts are for precision  $10^{-10}$  with optimal starting point.

**Remark 14.1.** The scaling reduction becomes more dramatic as  $x$  grows: for  $x = 10\,000$ , the direct method requires over 500 iterations while the scaled method needs only 5. This is because  $x' \rightarrow 1$  as  $x \rightarrow \infty$ , pushing  $\lambda_{p,x'}$  toward the regime  $\lambda \sim (p-1)(\alpha' - 1)/2 \ll 1$  of Proposition 9.1(1).

## 14.2 Geometric interpretation: Thales preconditioning

The scaling optimization admits an elegant geometric interpretation on Pandrosion’s original Thales diagram.

In the direct construction, one seeks to insert  $p-1$  proportional means in a segment of ratio  $x : 1$ . When  $x \gg 1$ , this amounts to constructing parallels in a very elongated rectangle, where successive intercepts move slowly toward the target — geometrically, the parallels are nearly horizontal.

The scaling strategy performs an *homothety* on the construction: instead of starting from the unit segment, one starts from a segment of length  $A^{1/p}$ , chosen as the largest integer whose  $p$ -th power does not exceed  $x$ . The remaining construction now fills a much smaller gap: the rectangle has ratio  $x' = x/A$  close to 1, and the parallels converge rapidly.

**Proposition 14.2** (Double preconditioning). The scaling method combines two distinct acceleration mechanisms:

1. **Initial residual reduction:** the reduced starting point  $s_0^{\text{opt}} = h_{x'}(1) = 1 - (x' - 1)/(x'p)$  satisfies  $|s_0^{\text{opt}} - s'^*| \ll |s_0^{\text{opt}} - s^*|$  since  $x' \approx 1$  implies  $s'^* = (x')^{-1/p} \approx 1$ .
2. **Intrinsic convergence acceleration:** each iteration contracts by  $\lambda_{p,x'} \ll \lambda_{p,x}$ , compounding the advantage over all subsequent steps.

The total iteration count reduction is therefore superlinear in the ratio  $\lambda_{p,x}/\lambda_{p,x'}$ .

*Proof.* For mechanism (1):  $\varepsilon'_0 = |v'_0 - \alpha'| \leq C \cdot |x' - 1| \rightarrow 0$  as  $x' \rightarrow 1$ , compared to  $\varepsilon_0 = |v_0 - \alpha| \geq C' \cdot |x - 1| \gg 1$  for the unscaled problem.

For mechanism (2): the iteration count satisfies

$$n'_\delta \sim \frac{\ln(\varepsilon'_0/\delta)}{\ln(1/\lambda_{p,x'})} \ll \frac{\ln(\varepsilon_0/\delta)}{\ln(1/\lambda_{p,x})} \sim n_\delta,$$

since both the numerator decreases (smaller  $\varepsilon'_0$ ) and the denominator increases (smaller  $\lambda_{p,x'}$ , hence larger  $\ln(1/\lambda_{p,x'})$ ). These two effects multiply.  $\square$

**Remark 14.2** (From Pandrosion to a practical algorithm). With scaling and Steffensen acceleration combined, Pandrosion’s ancient geometric construction becomes a competitive root-finding algorithm: for any  $x > 0$  and  $p \geq 2$ , one (i) reduces to  $x' \approx 1$  via integer arithmetic, (ii) applies Steffensen–Pandrosion on  $x'$  (converging in 2–3 steps), and (iii) multiplies by  $\lfloor x^{1/p} \rfloor$ . The total cost is 4–6 evaluations of  $S_p$  plus one integer  $p$ -th root — a purely algebraic, derivative-free computation whose origins lie in a ruler-and-parallels construction from fourth-century Alexandria.

### 14.3 Multi-precision benchmark against Newton and Halley

We close Part I with an experimental sanity check: how does the full Steffensen–Pandrosion–scaling pipeline compare against textbook Newton and Halley iterations for  $x^{1/p}$  at high precision? The script `latex/benchmark_multiprec.py` runs the three methods in MPFR-backed arithmetic (via the `mpmath` library) at precision targets of 50, 200, and 500 decimal digits, starting all three methods from the same integer seed  $\lfloor x^{1/p} \rfloor$  and stopping on the same condition  $|u^p - x|/x \leq 10^{-\text{digits}}$ . Table 11 summarises the iteration counts; wall-clock times are reported in the repository.

| Regime                   | 50 digits |   |    | 200 digits |   |    | 500 digits |   |    |
|--------------------------|-----------|---|----|------------|---|----|------------|---|----|
|                          | N         | H | SP | N          | H | SP | N          | H | SP |
| $p = 2, x = 2$           | 7         | 4 | 5  | 9          | 6 | 7  | 10         | 6 | 8  |
| $p = 3, x = 2$           | 7         | 4 | 5  | 9          | 6 | 7  | 10         | 6 | 8  |
| $p = 3, x = 11$          | 6         | 4 | 4  | 8          | 5 | 6  | 10         | 6 | 7  |
| $p = 3, x = 10^6 + 3$    | 4         | 2 | 2  | 6          | 4 | 4  | 7          | 5 | 5  |
| $p = 5, x = 99$          | 8         | 5 | 5  | 10         | 6 | 7  | 11         | 7 | 9  |
| $p = 10, x = 2^{20} + 3$ | 4         | 2 | 2  | 6          | 4 | 4  | 7          | 5 | 5  |
| $p = 20, x = 2$          | 7         | 4 | 5  | 9          | 6 | 7  | 11         | 6 | 8  |

**Table 11:** Iteration counts to reach the stated precision, starting from  $u_0 = \lfloor x^{1/p} \rfloor$ . **N** = Newton, **H** = Halley, **SP** = Steffensen–Pandrosion with scaling (Proposition 14.1). All three methods exhibit  $\mathcal{O}(\log \log(1/\varepsilon))$  growth in iteration count; going from 50 to 500 decimal digits (a  $10\times$  increase in precision exponent) adds about 3–4 iterations to each method, matching the theoretical  $\log_2 \log$  scaling.

#### Observations.

1. **Iteration count scaling.** All three methods grow as  $\log_2 \log(1/\varepsilon)$ , as predicted by theory (Newton and Steffensen–Pandrosion are order 2; Halley is order 3). In the table, going from 50 to 500 digits adds  $\approx 3$ –4 iterations, consistent with  $\log_2 \log_{10}(10^{500}/10^{50}) = \log_2(500/50) \approx 3.3$ .
2. **Halley wins on count.** Halley (cubic order) consistently takes 2–3 fewer iterations than Newton (quadratic order), saving roughly 30% on iteration count. Newton and Steffensen–Pandrosion–scaling differ by at most 1–2 iterations in every regime.
3. **Wall clock.** Per-iteration, Newton is fastest (one power  $u^{p-1}$  per step), Halley is close to Newton (one power  $u^p$  and a rational combination), and Steffensen–Pandrosion–scaling is 2–3 $\times$  slower per iteration than Halley because each Aitken step requires two evaluations of the degree- $p$  polynomial  $S_p$ . On the iteration counts above and measured in the repository, Halley is the fastest method in total wall-clock time, Newton is within 10–30%, and Steffensen–Pandrosion–scaling sits 1.5–3 $\times$  slower.
4. **The role of scaling.** Without scaling, Steffensen–Pandrosion on  $x = 10^6 + 3$  with  $p = 3$  would be severely handicapped by the contraction ratio  $\lambda_{3,10^6+3} \approx 0.9997$ : the unaccelerated Pandrosion iteration would need  $\lceil \ln(10^{-200})/\ln(1/\lambda) \rceil \approx 1.5 \times 10^6$  iterations

to reach 200-digit precision. The Aitken upgrade reduces this dramatically (to  $\mathcal{O}(\log \log)$ ), but its constant  $K_S \sim |h''(s^*)|/(2(1-\lambda))$  is inflated by  $1/(1-\lambda) \approx 3300$ , delaying the entry into the quadratic regime. With scaling ( $x' \approx 1$ ,  $\lambda_{3,x'}$  small), both issues disappear and the count collapses to 4. The scaling preconditioning is what makes Steffensen–Pandrosion competitive at large  $x$ .

**Conclusion of the benchmark.** Steffensen–Pandrosion–scaling is not the fastest  $p$ -th root method in practice: Halley wins on both iteration count and wall clock, and Newton wins on wall clock by a small margin. What the benchmark does establish is that the scheme — with scaling — is *competitive*: it achieves the same  $\mathcal{O}(\log \log)$  asymptotic as Newton and Halley, within a small constant factor of them in every regime tested, while being derivative-free. For applications where the derivative is costly or unavailable, Steffensen–Pandrosion–scaling is a viable alternative; for standard  $p$ -th root computation, Halley remains the method of choice.

## Part II

# Complex extension and formal verification

Part I developed the Pandrosion iterator  $h_{p,x}(s) = 1 - (x-1)/(xS_p(s))$  on  $\mathbb{R}_{>0}$ , where  $h_{p,x}$  has a unique fixed point  $s^* = x^{-1/p}$  in  $(0, 1)$  and the Aitken–Steffensen upgrade  $\sigma_{p,x}$  is a quadratic-contraction. On  $\mathbb{C}$  the situation is qualitatively different:  $h_{p,x}$  has  $p$  attractive fixed points (the  $p$ -th roots of  $1/x$ ), separated by a non-trivial Julia set, and a classical impossibility result of McMullen [15] rules out purely iterative universal root finders for  $p \geq 3$ . Part II extends the analysis to this complex setting via a multi-start scheme whose seed is target-dependent — an  $\alpha$ -dependent preconditioning that takes the method out of the hypotheses of [15].

**Structural warning.** The scheme of Part II requires a *target anchor*  $\alpha$  (one of the  $p$  solutions of  $z^p = 1/x$ ) to be chosen before the seed is constructed. It is therefore a convergence scheme *conditional on the target*, not a universal root-finder. Readers who expect the latter will misread Theorem 16.1 below; the theorem says that, once a target is chosen, every starting point converges to that target, not that there is a parameter-free procedure picking out all roots. This is the unavoidable price of side-stepping McMullen’s barrier.

**From the real to the complex quadratic bound.** The Aitken–Steffensen map  $\sigma_{p,x}$  is rational in  $z$ , so holomorphic on  $\mathbb{C}$  away from the finite set  $\text{sing}(\sigma_{p,x}) := \{z : S_p(z) = 0 \text{ or } h_{p,x}(h_{p,x}(z)) - 2h_{p,x}(z) + z = 0\}$ . At any cyclotomic fixed point  $\alpha$  (i.e.  $\alpha^p = 1/x$  with  $S_p(\alpha) \neq 0$ ), the factorization of Part I gives  $\sigma_{p,x}(\alpha) = \alpha$  and  $\sigma'_{p,x}(\alpha) = 0$ .

We now cascade three quantities to avoid the apparent circularity between  $K_\sigma$  and  $R_\sigma$ .

*Step 1 fix a reference radius.* Set

$$R_0 := \frac{1}{2} \text{dist}(\alpha, \text{sing}(\sigma_{p,x})),$$

so that  $\overline{B}(\alpha, R_0) \subset \mathbb{C} \setminus \text{sing}(\sigma_{p,x})$  and  $\sigma_{p,x}$  is holomorphic on this closed disk.

*Step 2 explicit quadratic constant via Cauchy.* Define

$$M_0 := \sup_{|w-\alpha|=R_0} |\sigma_{p,x}(w) - \alpha|, \quad K_\sigma := M_0/R_0^2.$$

By Cauchy’s estimate applied to the holomorphic function  $\sigma_{p,x}(\cdot) - \alpha$ , which vanishes to order 2 at  $\alpha$  (since  $\sigma_{p,x}(\alpha) = \alpha$  and  $\sigma'_{p,x}(\alpha) = 0$ ), we obtain

$$|\sigma_{p,x}(z) - \alpha| \leq \frac{M_0}{R_0^2} |z - \alpha|^2 = K_\sigma |z - \alpha|^2 \quad \text{for } z \in \overline{B}(\alpha, R_0).$$



*Step 3 admissible contraction radius.* Define

$$R_\sigma = R_\sigma(x, p, \alpha) := \min(R_0, 1/(2K_\sigma)).$$

Then  $R_\sigma \leq R_0$  (so the Cauchy bound of Step 2 is valid on  $\overline{B}(\alpha, R_\sigma)$ ) and  $K_\sigma R_\sigma \leq 1/2$ . All three quantities  $R_0, K_\sigma, R_\sigma$  are explicit functions of  $(x, p, \alpha)$ , computed in cascade without circularity. The seed construction below uses  $K_\sigma$  and  $R_\sigma$  as primitive quantities.

## 15 Cyclotomic anchors and the complex iterator

**Definition 15.1** (Cyclotomic anchors). For  $\alpha \in \mathbb{C}^*$  and  $p \geq 1$ , let

$$\gamma_k := \alpha \cdot e^{2\pi i k/p}, \quad k = 0, 1, \dots, p-1.$$

When  $\alpha^p = 1/x$ , the  $\gamma_k$  are the  $p$  distinct solutions of  $z^p = 1/x$ . By direct expansion of Definition 15.1,  $\omega \cdot \gamma_k = \gamma_{(k+1) \bmod p}$  (with  $\omega = e^{2\pi i/p}$ ),  $\sum_k \gamma_k = 0$ , and  $\prod_k \gamma_k = (-1)^{p-1}x$ . These are definitional consequences for  $z^p = x$ , not specializations of the classical Galois or Vieta theorems; we use them freely.

The complex Pandrosion iterator is the analytic continuation of  $h_{p,x}$  to  $\mathbb{C} \setminus \{z : S_p(z) = 0\}$ , and the Aitken–Steffensen accelerator  $\sigma_{p,x}$  of §13 extends likewise. The quadratic local bound of Part I applies verbatim to  $\sigma_{p,x}$  in a disk  $B(\gamma_k, R_\sigma)$  around every cyclotomic anchor, with the same explicit constants  $K_\sigma$  and  $R_\sigma$  satisfying  $K_\sigma R_\sigma \leq 1/2$ .

## 16 Universal multi-start convergence in $\mathbb{C}$

### 16.1 The multi-start seed

**Definition 16.1** (Multi-start seed). For a cyclotomic anchor  $\alpha \in \mathbb{C}^*$  with  $\alpha^p = 1/x$  and  $z \in \mathbb{C}$ ,

$$\text{seed}_{\alpha, \varepsilon}(z) := \alpha + \varepsilon \cdot (z - \alpha), \quad \varepsilon := \frac{R_\sigma(x, p, \alpha)}{2 \|z - \alpha\|} \quad (\varepsilon := 1 \text{ if } z = \alpha), \quad (17)$$

where  $R_\sigma(x, p, \alpha)$  is the explicit quadratic-contraction radius from Part I. The seed always lies in  $\overline{B}(\alpha, R_\sigma/2)$ .

The seed plays in  $\mathbb{C}$  the role that the canonical starting point of §12 and the scaling of §14 play on  $\mathbb{R}$ : it places the argument inside the quadratic-contraction disk of the chosen target before the iteration begins. It is  $\alpha$ -dependent by design, and this dependency is what moves the scheme out of the hypotheses of [15].

### 16.2 Universal multi-start theorem

**Theorem 16.1** (Universal multi-start convergence). Let  $x \in \mathbb{C}^*$ ,  $p \geq 1$ ,  $\alpha \in \mathbb{C}^*$  with  $\alpha^p = 1/x$  and  $S_p(\alpha) \neq 0$ , and let  $z \in \mathbb{C}$  with  $z \neq \alpha$ . Define the seed parameter

$$\varepsilon_{\text{seed}} := \frac{R_\sigma(x, p, \alpha)}{2 \|z - \alpha\|},$$

which depends only on  $(x, p, \alpha, z)$ . Then for every target accuracy  $\delta > 0$ , the integer  $N := N_{\text{eff}}(K_\sigma, R_\sigma, \delta)$  of Theorem 16.2 satisfies

$$\forall n \geq N : \quad \|\sigma_{p,x}^n(\text{seed}_{\alpha, \varepsilon_{\text{seed}}}(z)) - \alpha\| \leq \delta.$$

The case  $z = \alpha$  is trivial (take  $\varepsilon_{\text{seed}} := 1$  and  $N := 0$ ). The construction separates cleanly into two levels: the seed parameter  $\varepsilon_{\text{seed}}$  and the admissible radius  $R_\sigma$  are functions of  $(x, p, \alpha, z)$  alone, while the iteration count  $N$  alone depends on the target accuracy  $\delta$ .



*Proof.* Let  $d := \|z - \alpha\|$ ; we may assume  $d > 0$  (else  $z = \alpha$  is already the target). With  $\varepsilon_{\text{seed}} := R_\sigma/(2d)$ , set

$$z_0 := \text{seed}_{\alpha, \varepsilon_{\text{seed}}}(z) = \alpha + \varepsilon_{\text{seed}}(z - \alpha).$$

Then  $\|z_0 - \alpha\| = \varepsilon_{\text{seed}} \cdot d = R_\sigma/2 < R_\sigma$ , so  $z_0 \in B(\alpha, R_\sigma)$ .

The quadratic-contraction bound of Part I (whose proof applies verbatim to  $\sigma_{p,x}$  viewed as an analytic map on any small enough disk centred at a fixed point, since  $\sigma_{p,x}(\alpha) = \alpha$  and  $\sigma'_{p,x}(\alpha) = 0$ ) reads: for  $w \in B(\alpha, R_\sigma)$ ,

$$\|\sigma_{p,x}(w) - \alpha\| \leq K_\sigma \|w - \alpha\|^2,$$

with  $K_\sigma R_\sigma \leq 1/2$ . Hence if  $w_n := \sigma_{p,x}^n(z_0)$  satisfies  $\|w_n - \alpha\| \leq R_\sigma$ , then  $\|w_{n+1} - \alpha\| \leq K_\sigma \|w_n - \alpha\|^2 \leq K_\sigma R_\sigma \cdot \|w_n - \alpha\| \leq \frac{1}{2} \|w_n - \alpha\| \leq R_\sigma/2$ , so the orbit stays in  $\overline{B}(\alpha, R_\sigma/2)$  by induction from  $w_0 = z_0$ .

Setting  $e_n := \|w_n - \alpha\|$ , one has  $e_0 \leq R_\sigma/2$  and  $e_{n+1} \leq K_\sigma e_n^2$ . Theorem 16.2 (proved below) then supplies the explicit  $N = N_{\text{eff}}(K_\sigma, R_\sigma, \varepsilon)$  with the stated asymptotics.  $\square$

**Remark 16.1** (The hypothesis  $S_p(\alpha) \neq 0$ ). The condition  $S_p(\alpha) \neq 0$  rules out the degenerate case where the denominator of  $h_{p,x}$  vanishes at the anchor. Since  $S_p(\alpha) = (\alpha^p - 1)/(\alpha - 1)$  for  $\alpha \neq 1$ , one has  $S_p(\alpha) = 0$  iff  $\alpha^p = 1$  with  $\alpha \neq 1$ , which combined with  $\alpha^p = 1/x$  forces  $x = 1$  and  $\alpha$  a non-trivial  $p$ -th root of unity. The hypothesis therefore fails on a single point of the  $x$ -line, and holds for every  $x \neq 1$ .

### 16.3 Explicit iteration count

**Definition 16.2** (Effective iteration count). For  $K, R, \varepsilon > 0$  with  $KR < 1$  and  $K\varepsilon < 1$ ,

$$N_{\text{eff}}(K, R, \varepsilon) := \log_2 \lceil \log(K\varepsilon)/\log(KR) \rceil_+ + 1,$$

with  $\lceil \cdot \rceil_+$  the natural-number ceiling clamped at 0.

**Theorem 16.2** (Explicit iteration count). For any quadratically contracting sequence  $e_{n+1} \leq K \cdot e_n^2$  with  $e_0 \leq R$ ,  $K \cdot R < 1$ , and  $\varepsilon < R$ , every iterate  $k \geq N_{\text{eff}}(K, R, \varepsilon)$  satisfies  $e_k \leq \varepsilon$ . Asymptotically  $N_{\text{eff}} \sim \log_2 \log(1/\varepsilon)$ .

*Proof.* Set  $u_n := Ke_n$ , so  $u_0 \leq KR < 1$  and  $u_{n+1} \leq u_n^2$ . By induction  $u_n \leq u_0^{2^n} \leq (KR)^{2^n}$ . We want  $u_n \leq K\varepsilon$ , i.e.  $(KR)^{2^n} \leq K\varepsilon$ . Since  $KR < 1$  and  $K\varepsilon < 1$ , both  $\log(KR)$  and  $\log(K\varepsilon)$  are negative; dividing by  $\log(KR) < 0$  reverses the inequality direction and gives  $2^n \geq \log(K\varepsilon)/\log(KR)$ , i.e.

$$n \geq \log_2 \left\lceil \frac{\log(K\varepsilon)}{\log(KR)} \right\rceil_+,$$

which is achieved at  $n = N_{\text{eff}}(K, R, \varepsilon)$  by the +1 safety margin. The asymptotics  $N_{\text{eff}} \sim \log_2 \log(1/\varepsilon)$  as  $\varepsilon \rightarrow 0$  follow since  $\log(K\varepsilon)/\log(KR) \sim \log(1/\varepsilon)/|\log(KR)|$ .  $\square$

### 16.4 Arbitrary target selector

**Theorem 16.3** (Target selection). For every function  $S : \mathbb{C} \rightarrow \text{Fin } p$ , every  $z \in \mathbb{C}$ , and every  $\varepsilon > 0$ , the multi-start scheme with target  $\gamma_{S(z)}$  converges to  $\gamma_{S(z)}$  with an iteration count of the form of Theorem 16.2.

Three selectors are formalized: constant (`principal_selector`), Voronoï-nearest (`nearest_anchor_selector`), angular sector.

## 16.5 Multi-start compatibility bundle

The previous subsections presented the ingredients of the multi-start pipeline separately: the seed (Definition 16.1), the universal convergence theorem (Theorem 16.1), the target selectors (Theorem 16.3), and the real-line preconditionings of Part I transported to the cyclotomic anchor family. The following statement bundles all five into a single compatibility assertion. It is a glue lemma used by downstream citations, not a headline result.

**Theorem 16.4** (Multi-start compatibility bundle). Let  $x \in \mathbb{C}^*$ ,  $A \in \mathbb{C}^*$ ,  $p \geq 1$ , and  $\alpha, \beta \in \mathbb{C}^*$  with  $\alpha^p = x$  and  $\beta^p = A$ . Let  $\gamma : \{0, 1, \dots, p-1\} \rightarrow \mathbb{C}^*$  be a cyclotomic anchor family satisfying  $\gamma_k^p = A/x$  and  $S_p(\gamma_k) \neq 0$  for every  $k$ . Then for every query point  $z_0 \in \mathbb{C}$ , the following five properties hold simultaneously:

- (1) **Multi-start target selection.** There exists an index  $s^* \in \{0, 1, \dots, p-1\}$  — the Voronoï-nearest anchor to  $z_0$  — with  $\|\gamma_{s^*} - z_0\| \leq \|\gamma_t - z_0\|$  for every  $t$ .
- (2) **Steffensen fixing.** Every  $\gamma_s$  is a fixed point of the scaled Steffensen map:  $\sigma_{x/A, p}(\gamma_s) = \gamma_s$  for every  $s$ .
- (3) **Scaling compatibility.** The principal ratio satisfies  $(\alpha/\beta)^p = x/A$ .
- (4) **Reconstruction.**  $\alpha = \beta \cdot (\alpha/\beta)$ , so the original  $p$ -th root is recovered from the scaled one by multiplication by the integer root  $\beta$ .
- (5) **Optimal starting point.** The canonical seed for the scaled problem is  $s_0^{\text{opt}}(x/A) = h_{p, x/A}(1)$ , matching Proposition 12.1 of Part I.

*Proof.* (1) Voronoï-nearest existence: the function  $t \mapsto \|\gamma_t - z_0\|$  attains its minimum on the finite set  $\{0, \dots, p-1\}$  at some index  $s^*$ .

(2) Steffensen fixing: by the factorization of Part I specialized to  $\gamma_s^p = A/x = 1/(x/A)$ ,  $h_{p, x/A}(\gamma_s) = \gamma_s$ ; applying Aitken's  $\Delta^2$  to a constant sequence  $h(\gamma_s), h(h(\gamma_s)), \dots$  yields  $\sigma_{x/A, p}(\gamma_s) = \gamma_s$  whenever the denominator of the Aitken step is non-zero, which holds since  $S_p(\gamma_s) \neq 0$  (the removable singularity at a fixed point is handled by the standard Aitken convention).

(3) Scaling compatibility:  $(\alpha/\beta)^p = \alpha^p/\beta^p = x/A$  by direct computation.

(4) Reconstruction:  $\beta \cdot (\alpha/\beta) = \alpha$  since  $\beta \neq 0$ .

(5) Optimal starting point:  $h_{p, x/A}(1) = 1 - ((x/A) - 1)/((x/A) \cdot S_p(1)) = 1 - ((x/A) - 1)/((x/A) \cdot p)$ , matching (13) applied to the scaled problem  $x' = x/A$ .  $\square$

**Remark 16.2** (What the compatibility bundle actually says). Theorem 16.4 is not itself a new mathematical result; each of the five parts is proven elsewhere in Parts I and II. Its value is structural: it records, in a single statement, the *compatibility* of the five layers of the pipeline — selection, fixing, scaling, reconstruction, seeding — so that downstream statements (in the formalization and in the effective iteration-count theorem below) can cite one name instead of five. The theorem's role is that of a *glue lemma*: Proposition (3), for instance, is used by the scaling preconditioning of Part I to certify that a  $\mathbb{R}$ -level identity transports correctly to the cyclotomic family in  $\mathbb{C}$ .

**Remark 16.3** (Effective version). The effective counterpart of Theorem 16.4 replaces the asymptotic iteration count  $N$  of Theorem 16.1 with the explicit  $N_{\text{eff}}(K_\sigma, R_\sigma, \delta)$  of Theorem 16.2, applied per-anchor in the family  $\gamma$ . Concretely, for every  $z_0 \in \mathbb{C}$  and every target accuracy  $\delta > 0$ , the following holds: choose  $s^*$  as in Theorem 16.4(1), form the seed  $\text{seed}_{\gamma_{s^*}, \varepsilon_{\text{seed}}}(z_0)$  with  $\varepsilon_{\text{seed}} = R_\sigma(x/A, p, \gamma_{s^*})/(2\|z_0 - \gamma_{s^*}\|)$ , and iterate  $\sigma_{x/A, p}$ . After  $N = N_{\text{eff}}(K_\sigma, R_\sigma, \delta)$  steps, the iterate is within  $\delta$  of  $\gamma_{s^*}$ , and multiplying by  $\beta$  yields an approximation of  $\alpha$  to comparable accuracy. The total cost is  $\log_2 \log(1/\delta) + O(1)$  iterations plus one integer  $p$ -th root — the scaling cost  $\beta$  — plus one final multiplication, exactly the pipeline the benchmark of §14.3 exercises.

## 16.6 Explicit lower bound on the basin of attraction (unseeded iteration)

Theorem 16.1 concerns the *seeded* iteration: every starting point  $z$  is first mapped via  $\text{seed}_{\alpha, \varepsilon_{\text{seed}}}$  into the quadratic-contraction disk before  $\sigma_{p,x}$  is applied. We now record a complementary, *unconditional* statement about the *unseeded* iteration: even without any preprocessing, there is an explicit disk around each cyclotomic anchor on which plain iteration of  $\sigma_{p,x}$  already converges. This gives an explicit lower bound on the volume of the (true, dynamical) basin of attraction of each anchor.

**Definition 16.3** (Basin of attraction). For a cyclotomic anchor  $\gamma_k$  of  $\sigma_{p,x}$ , set

$$\Omega_k := \{z \in \mathbb{C} : \sigma_{p,x}^n(z) \text{ is defined for all } n \geq 0 \text{ and } \sigma_{p,x}^n(z) \rightarrow \gamma_k \text{ as } n \rightarrow \infty\}.$$

**Proposition 16.1** (Explicit basin lower bound). Under the standard hypotheses on  $(x, p, \gamma_k)$  (namely  $\gamma_k^p = 1/x$  and  $S_p(\gamma_k) \neq 0$ ), the disk  $\overline{B}(\gamma_k, R_\sigma(x, p, \gamma_k))$  lies entirely in  $\Omega_k$ . Consequently:

- (i) **Per-anchor volume:**  $\text{vol}(\Omega_k) \geq \pi R_\sigma(x, p, \gamma_k)^2$ .
- (ii) **Disjointness:** the basins  $\Omega_0, \dots, \Omega_{p-1}$  are pairwise disjoint.

$$(iii) \text{ **Total volume:} \text{vol}\left(\bigcup_{k=0}^{p-1} \Omega_k\right) \geq \sum_{k=0}^{p-1} \pi R_\sigma(x, p, \gamma_k)^2.**$$

*Proof.* Fix  $k$  and abbreviate  $R := R_\sigma(x, p, \gamma_k)$ ,  $K := K_\sigma(x, p, \gamma_k)$ . From the three-step construction in Part II's opening, we have  $KR \leq 1/2$  and

$$|\sigma_{p,x}(z) - \gamma_k| \leq K |z - \gamma_k|^2 \quad \text{for all } z \in \overline{B}(\gamma_k, R_0),$$

with  $R \leq R_0$ , so the bound holds a fortiori on  $\overline{B}(\gamma_k, R)$ .

*Forward invariance.* For  $z \in \overline{B}(\gamma_k, R)$ ,

$$|\sigma_{p,x}(z) - \gamma_k| \leq K |z - \gamma_k|^2 \leq (KR) |z - \gamma_k| \leq \frac{1}{2} |z - \gamma_k| \leq R/2,$$

hence  $\sigma_{p,x}(\overline{B}(\gamma_k, R)) \subset \overline{B}(\gamma_k, R/2) \subset \overline{B}(\gamma_k, R)$ .

*Convergence.* Applying the inequality above inductively,  $|\sigma_{p,x}^n(z) - \gamma_k| \leq 2^{-n} |z - \gamma_k| \leq 2^{-n} R \rightarrow 0$ , so  $z \in \Omega_k$ . This proves the disk containment.

Part (i) follows since  $\overline{B}(\gamma_k, R) \subset \Omega_k$  gives  $\text{vol}(\Omega_k) \geq \pi R^2$ .

Part (ii) is immediate: if  $z \in \Omega_j \cap \Omega_k$  with  $j \neq k$ , then  $\sigma_{p,x}^n(z)$  would converge simultaneously to  $\gamma_j$  and  $\gamma_k$ , which is impossible since  $\gamma_j \neq \gamma_k$ .

Part (iii) follows from (i) and (ii) by additivity of Lebesgue measure.  $\square$

**Remark 16.4** (Comparison with Theorem 16.1). Theorem 16.1 covers every  $z \in \mathbb{C}$  but only after pre-processing  $z$  into a chosen target's disk via the seed. Proposition 16.1 does not need any pre-processing: the disk  $\overline{B}(\gamma_k, R_\sigma)$  is directly in the unseeded basin. The two results are complementary: the seed (Theorem 16.1) addresses points outside any anchor's disk, while the basin bound (Proposition 16.1) quantifies the unconditional contribution from points already close to an anchor.

**Remark 16.5** (Tightness and numerical observations). The bound  $\pi R_\sigma^2$  is typically very loose. Numerical experiments (script `latex/basin_estimate.py`) on  $(p, x) = (3, 2)$  show that  $\Omega_0$  (principal anchor) covers approximately 99.95% of the grid  $[-1.5, 1.5]^2$ , while the analytic lower bound  $\pi R_\sigma^2$  with an empirical  $R_\sigma \geq 2$  gives  $\text{vol}(\Omega_0) \geq 4\pi \approx 12.57$ . For  $(p, x) = (7, 10)$  the radius tightens to  $R_\sigma \approx 0.37$  empirically, showing that the geometric obstruction of a non-trivial Julia boundary begins to bite at larger  $p$  and  $x$ . Proposition 16.1 is an explicit *lower* bound only; improving it to a matching upper bound or a Hausdorff-dimension estimate on  $\mathbb{C} \setminus \bigcup_k \Omega_k$  requires Fatou–Julia machinery not in Mathlib v4.7.0 and is left for future work.

**Remark 16.6** (Pullback improvement). The bound can be sharpened without leaving the elementary toolbox. Since  $\sigma_{p,x}$  is a rational map, each iterate  $\sigma_{p,x}^{-m}(\overline{B}(\gamma_k, R/2))$  is an open subset of  $\Omega_k$  (pullback of a forward-invariant open set containing  $\gamma_k$ ). Hence

$$\text{vol}(\Omega_k) \geq \text{vol}\left(\bigcup_{m \geq 0} \sigma_{p,x}^{-m}(\overline{B}(\gamma_k, R/2))\right),$$

and each pullback-level  $m$  contributes a disjoint (up to measure-zero overlap) additional region. We do not pursue this refinement here because each pullback involves solving a degree- $d$  polynomial equation, and the bookkeeping obscures the elementary character of the  $m = 0$  bound.

## 16.7 Multi-precision benchmark in the complex plane

Complementing the real-line benchmark of §14.3, the script `latex/benchmark_complex.py` compares four methods on the complex problem  $z^p = X$  for  $X \in \mathbb{C}^*$ :

- Newton’s iteration  $z_{n+1} = ((p-1)z_n + X/z_n^{p-1})/p$ ;
- Halley’s iteration  $z_{n+1} = z_n \cdot ((p-1)z_n^p + (p+1)X)/((p+1)z_n^p + (p-1)X)$ ;
- Multi-start Steffensen–Pandrosion (Theorem 16.1), seeded at the principal anchor  $\alpha = X^{1/p}$ ;
- Scaled multi-start (the compatibility bundle of Theorem 16.4), combining scaling  $\beta = \lfloor |X|^{1/p} \rfloor$ , Voronoi-nearest selector on the anchor family, seed, and reconstruction.

All methods start from the same user-supplied  $z_0 \in \mathbb{C}$  and stop when  $|z_n^p - X|/|X| \leq 10^{-d}$  at precision target  $d$  decimal digits. The seed radius is set uniformly to a conservative  $R_\sigma = 0.5$ .

| Regime                                  | 50 digits |   |    |          | 200 digits |   |    |          |
|-----------------------------------------|-----------|---|----|----------|------------|---|----|----------|
|                                         | N         | H | MS | GM       | N          | H | MS | GM       |
| $p = 3, X = 2 + 3i, z_0 = 1$            | 8         | 5 | 7  | 7        | 10         | 6 | 9  | 9        |
| $p = 3, X = 10, z_0 = 2 + i$            | 8         | 5 | 7  | 6        | 10         | 6 | 9  | 8        |
| $p = 4, X = -1 + i, z_0 = 1 + i$        | 8         | 5 | 8  | 8        | 10         | 6 | 10 | 10       |
| $p = 5, X = 100 + 7i, z_0 = 2 + i$      | 12        | 6 | 8  | 7        | 14         | 7 | 10 | 9        |
| $p = 3, X = 10^6 + 3, z_0 = 50$ (far)   | 9         | 5 | 6  | <b>4</b> | 11         | 6 | 8  | <b>6</b> |
| $p = 3, X = 10^6 + 17i, z_0 = 50 + 10i$ | 9         | 5 | 6  | <b>4</b> | 11         | 6 | 8  | <b>6</b> |
| $p = 7, X = 10, z_0 = 1 + i$            | 7         | 4 | 8  | 13       | 9          | 6 | 10 | 15       |

**Table 12:** Complex-case iteration counts. **N** = Newton, **H** = Halley, **MS** = multi-start Pandrosion seeded at principal anchor, **GM** = scaled multi-start (compatibility-bundle pipeline: scaling + Voronoi-nearest + seed). Stopping criterion  $|z^p - X|/|X| \leq 10^{-d}$ . Boldface marks the best (or tied-best) method in each row; Halley is best-or-tied in most regimes, scaled multi-start wins on large- $|X|$  regimes where scaling collapses the problem to  $X' \approx 1$ .

### Observations.

1. **Halley is the iteration-count winner.** Cubic order gives Halley a consistent 30–40% edge on iteration count in every tested regime. Wall-clock ordering is the same as the real case (Halley < Newton < MS < GM).
2. **Multi-start Pandrosion matches Newton within  $\pm 2$  iterations.** The seed construction places  $z_0$  inside the quadratic-contraction disk of the principal anchor, from which the  $\log_2 \log$  scaling kicks in. The iteration count is within one or two of Newton’s across all regimes.

3. **Scaled multi-start wins when scaling collapses the problem.** For  $|X| \gtrsim 10^5$ , the scaled multi-start drops to 4–6 iterations (at 50 and 200 digits respectively), a 30–40% improvement over plain multi-start Pandrosion, because  $X/\beta^p$  is close to 1 and  $\lambda_\infty(X/\beta^p)$  is small. This confirms experimentally the scaling preconditioning of Part I (Proposition 14.1) in the complex setting.
4. **Scaled multi-start can be worse in niche regimes.** For  $p = 7, X = 10, z_0 = 1 + i$ , it takes 13–15 iterations versus 8–10 for plain multi-start. The cause is that  $|X|^{1/7} \approx 1.39 < 2$ , so  $\beta = 1$  provides no preconditioning, yet the Voronoï-nearest selector picks a non-principal anchor for which the conservative  $R_\sigma = 0.5$  is not tight, forcing extra pre-asymptotic iterations.
5. **All four methods achieve  $\mathcal{O}(\log \log)$  scaling.** Going from 50 to 200 digits adds roughly 2–3 iterations to each method, matching  $\log_2(200/50) = 2$ .

**Bottom-line (complex case).** As on the real line, Steffensen–Pandrosion-style methods are *competitive* with textbook Newton/Halley at high precision: they remain within a small constant factor of Halley in every tested regime while being derivative-free, and the scaled multi-start pipeline provides an additional 30–40% savings in the large- $|X|$  regime where integer scaling bites. For derivative-cheap settings, Halley remains the fastest; for derivative-expensive or derivative-unavailable settings, the Pandrosion pipeline is a viable alternative.

## 17 Measurability and a.e. continuity of the solution map

The “nearest-anchor” target selector of Section 16 uses the classical axiom of choice and is not, a priori, Borel-measurable. We record a canonical Borel version and the elementary analytic properties it enjoys.

**Definition 17.1** (Borel selector and Voronoï boundary). Given anchors  $\gamma_0, \dots, \gamma_{p-1} \in \mathbb{C}$  (all distinct), define, for each  $z \in \mathbb{C}$ ,

$$S_{\text{Borel}}(z) := \min\{k \in \{0, \dots, p-1\} : \|\gamma_k - z\| = \min_j \|\gamma_j - z\|\},$$

breaking ties by smallest index. The *Voronoï boundary* of the anchor set is

$$\text{VorBdy} := \{z \in \mathbb{C} : \exists j \neq k, \|\gamma_j - z\| = \|\gamma_k - z\|\} = \bigcup_{j < k} B_{jk},$$

where  $B_{jk} := \{z : \|\gamma_j - z\| = \|\gamma_k - z\|\}$  is the perpendicular bisector of  $\gamma_j, \gamma_k$ . Let  $\text{VorInt} := \mathbb{C} \setminus \text{VorBdy}$ .

**Proposition 17.1** (Analytic structure of the solution map). The Borel solution map  $\Phi : z \mapsto \gamma_{S_{\text{Borel}}(z)}$  satisfies:

- (i) **Borel-measurability.**  $\Phi$  is Borel-measurable  $\mathbb{C} \rightarrow \mathbb{C}$ .
- (ii) **Null boundary.**  $\text{VorBdy}$  has two-dimensional Lebesgue measure zero.
- (iii) **A.e. continuity.**  $\Phi$  is locally constant on  $\text{VorInt}$ , hence continuous there; combined with (ii),  $\Phi$  is continuous at Lebesgue-a.e.  $z \in \mathbb{C}$ .

*Proof.* (i) *Borel-measurability.* The preimage  $\{z : S_{\text{Borel}}(z) = k\}$  is

$$\left( \bigcap_{j \neq k} \{z : \|\gamma_k - z\| \leq \|\gamma_j - z\|\} \right) \cap \left( \bigcap_{j < k} \{z : \|\gamma_k - z\| < \|\gamma_j - z\|\} \right),$$

a finite Boolean combination of closed (resp. open) half-planes in  $\mathbb{C}$ , hence Borel. So  $S_{\text{Borel}}$  is Borel-measurable, and  $\Phi$  is its composition with  $k \mapsto \gamma_k$  (continuous, a fortiori Borel).

(ii) *Null boundary.* Squaring  $\|\gamma_j - z\| = \|\gamma_k - z\|$  and expanding in real coordinates yields the affine equation

$$B_{jk} = \{z \in \mathbb{C} : \operatorname{Re}(\overline{(\gamma_k - \gamma_j)} z) = \frac{1}{2}(|\gamma_k|^2 - |\gamma_j|^2)\},$$

a strict affine subspace of  $\mathbb{C} \cong \mathbb{R}^2$  (Hausdorff dimension 1). Each  $B_{jk}$  therefore has two-dimensional Lebesgue measure zero. A finite union of null sets is null, so  $\operatorname{vol}(\operatorname{VorBdy}) = 0$ .

(iii) *A.e. continuity.* Fix  $z_0 \in \operatorname{VorInt}$  and let  $k_0 := S_{\text{Borel}}(z_0)$ . By definition of  $\operatorname{VorInt}$ ,  $\|\gamma_{k_0} - z_0\| < \|\gamma_j - z_0\|$  for all  $j \neq k_0$ , so some  $\delta > 0$  satisfies  $\|\gamma_{k_0} - z_0\| + 2\delta \leq \|\gamma_j - z_0\|$  for  $j \neq k_0$ . By continuity of  $z \mapsto \|\gamma_j - z\|$ , there is  $\eta > 0$  such that on  $B(z_0, \eta)$  both quantities change by at most  $\delta$ , giving  $\|\gamma_{k_0} - z\| < \|\gamma_j - z\|$  for  $j \neq k_0$ , so  $S_{\text{Borel}}(z) = k_0$  and  $\Phi(z) = \gamma_{k_0}$ . Thus  $\Phi$  is locally constant, hence continuous, on  $\operatorname{VorInt}$ . Combined with (ii),  $\Phi$  is continuous at Lebesgue-a.e.  $z$ .  $\square$

For backwards reference we also expose the individual statements.

**Theorem 17.1** (Borel measurability). The map  $S_{\text{Borel}}$  is Borel-measurable; so is  $\Phi$ .

**Theorem 17.2** (Voronoi boundary is Lebesgue-null).  $\operatorname{vol}(\operatorname{VorBdy}) = 0$ .

**Theorem 17.3** (A.e. continuity).  $\Phi$  is continuous at every  $z \in \operatorname{VorInt}$ ; combined with Theorem 17.2,  $\Phi$  is continuous at Lebesgue-a.e.  $z$ .

Both theorems are direct consequences of Proposition 17.1, (ii) and (iii) respectively. Together with Theorem 16.1, they give the analytic character of the multi-start solution map: Borel-measurable everywhere, locally constant outside a Lebesgue-null set.

## 18 Lean 4 formalization: corpus structure and elementary identities

### 18.1 Corpus overview

The Lean development lives at [github.com/ivan-fr/poussiere\\_de\\_x](https://github.com/ivan-fr/poussiere_de_x) and targets Lean 4 with Mathlib v4.7.0. It comprises 121 modules with zero `sorry`; every load-bearing theorem referenced in this paper has an audited axiomatic closure reducing to the Lean whitelist `{propext, Classical.choice, Quot.sound}`. Three layers of verifiability are offered to a third-party reviewer:

- **Continuous integration.** A GitHub Actions workflow (`.github/workflows/lean-build.yml`) compiles the full corpus on every push; a green CI badge in the repository `README.md` certifies that the current `main` branch compiles cleanly, with no `sorry`.
- **Axiom audit file.** The file `AXIOMS.md` at the repository root contains the raw `#print axioms` output for every load-bearing theorem cited in this paper; for each, the dependency closure reduces to the Lean whitelist `{propext, Classical.choice, Quot.sound}`.
- **Local reproduction.** Incremental build `bash scripts/lean-incremental.sh` takes 60–90 seconds on a modern laptop; per-theorem verification is available via `#print axioms <theorem>` at the end of the relevant module.

The modules are organised in layers: foundations (geometric sums, fixed-point characterization, Voronoi basins, cyclotomic anchor definitions), core spine (explicit super-attractive rate, global log log convergence, real-line  $p = 2$  discharge), multi-start (universal convergence, target selection, iteration count, compatibility bundle of §16.5), measure/topology (Borel measurability, Voronoi-null frontier, a.e. continuity), and a small set of classical specializations (Banach 1922, Aitken–Steffensen, Kronecker 1857, Aberth–Ehrlich, Kung–Traub attainability at  $c = 3$ ).



**Statement-level Props that are NOT proven.** A single archive module `PandrosionOpenProblems.lean` contains named Lean Props whose proofs are *deliberately absent*: the conjectural upper bound of Kung–Traub for  $c \geq 3$ , McMullen’s 1987 impossibility theorem, Lehmer’s Mahler-measure conjecture, and Smale’s 17th problem. These are recorded as `axiom` or `Prop` declarations without accompanying proof terms. They are not used downstream by any load-bearing theorem of the corpus, and their presence in the repository should not be mistaken for formal verification of the corresponding results. A reader auditing the corpus can confirm this by checking that no `theorem` depending on these Props appears in the load-bearing chain. A future cleanup will move this archive module to a dedicated `archive` namespace to make the non-load-bearing status structurally explicit.

## 18.2 Scope statement on classical open problems

No claim of progress on classical open problems of polynomial root-finding is made. The corpus touches Kung–Traub ( $c \geq 3$ ) through a compliance interface, not a proof, and records McMullen 1987 as a named statement without in-Lean proof (Mathlib v4.7.0 lacks the Fatou–Julia machinery needed). Lehmer’s Mahler-measure conjecture and Smale’s 17th problem are explicitly out of scope: the Pandrosion cyclotomic anchors produce algebraic integers whose Mahler measure is easy to compute (trivially far from Lehmer’s extremal regime), and the multi-start  $\mathcal{O}(\log \log(1/\varepsilon))$  complexity applies to the single-monomial family  $z^p = x$ , not to Smale’s coefficient-uniform problem.

## Part III

# Cyclotomic cancellation and two technical supplements

Part III gathers a short cyclotomic-cancellation section that records the one identity on the cyclotomic anchor family actually used downstream, together with two technical supplements: a no-cycle theorem for the seeded Steffensen orbit, and a uniform-in- $(p, x)$  log log complexity bound on a compact  $x$ -regime. These are not headline results; they are recorded because they close small logical gaps and provide a quantitative analogue of the benchmark observations of §14.3.

## 19 Cyclotomic cancellation on the anchor family

Let  $P(z) = z^p - 1/x$  with  $x \in \mathbb{C}^*$ ,  $p \geq 2$ ; let  $\alpha_0, \dots, \alpha_{p-1}$  be its roots,  $\alpha := x^{-1/p}$  principal,  $\omega := e^{2\pi i/p}$ . Then  $\alpha_k = \alpha \omega^k$  and

$$P'(\alpha_k) = p\alpha_k^{p-1} = \frac{p}{x\alpha_k}, \quad \nu_k := \frac{1}{P'(\alpha_k)} = \frac{x\alpha_k}{p}.$$

**Proposition 19.1** (Cyclotomic cancellation). For every  $p \geq 2$  and every  $s \in \mathbb{Z}$ ,

$$\sum_{k=0}^{p-1} \alpha_k^s = \begin{cases} 0 & \text{if } p \nmid s, \\ p\alpha^s & \text{if } p \mid s. \end{cases}$$

*Proof.*  $\sum_k \alpha_k^s = \alpha^s \sum_{k=0}^{p-1} (\omega^s)^k$ . If  $p \mid s$ ,  $\omega^s = 1$  and the sum is  $p$ ; otherwise  $\omega^s \neq 1$  is a  $p$ -th root of unity and the geometric sum vanishes.  $\square$

The relevant consequences for this paper are:

$$(i) \sum_{k=0}^{p-1} \nu_k = \frac{x}{p} \sum_k \alpha_k = 0 \text{ for } p \geq 2 \text{ (case } s = 1).$$

$$(ii) \text{ Closed-form product: } \prod_{k=0}^{p-1} P'(\alpha_k) = \frac{p^p}{x^p \prod_k \alpha_k} = (-1)^{p-1} \frac{p^p}{x^{p-1}}, \text{ using } \prod_k \alpha_k = (-1)^{p+1}/x$$

by Vieta on  $z^p - 1/x$ .

(iii) For  $p \geq 3$ :  $\sum_k P'(\alpha_k)^{-2} = (x/p)^2 \sum_k \alpha_k^2 = 0$ , and similarly  $\sum_k P'(\alpha_k)^2 = 0$  (case  $s = 2$ ,  $p \geq 3$ ,  $p \nmid 2$ ). The three special values  $\sum_k 1$ ,  $\sum_k P'(\alpha_k)^{-2}$ ,  $\sum_k P'(\alpha_k)^2$  at  $s = 0, \pm 1$  respectively equal  $p, 0, 0$  — a “cyclotomic cancellation” of all non-constant moments.

These three items are the arithmetic ingredients used implicitly in Part II: (i) is the vanishing of the first residue sum that underlies the a.e.-continuity analysis, and (ii)–(iii) normalize constants in auxiliary bounds. Items (i)–(iii) have been cross-checked symbolically in arbitrary precision for  $p \in \{2, 3, 5, 7, 10\}$  and representative  $x$  (real and complex).

**Remark 19.1** (Multi-start compatibility bundle). Combining (i) with the Part II structural results: for  $x, p, \alpha$  under the standard hypotheses, (A)  $\sum_k \nu_k = 0$ , (B) the multi-start image lies in the cyclotomic anchor set, (C) the Voronoï-nearest target is unique a.e., (D) the Voronoï boundary is Lebesgue-null (Theorem 17.2), (E) the Borel solution map is continuous a.e. (Theorem 17.3). All five statements are separately proven above; the bundling is a compatibility observation, not a new theorem.

**Remark 19.2** (Symmetry of the cyclotomic Voronoï diagram). The rotation  $T_\omega : z \mapsto \omega z$  is a Euclidean isometry permuting  $\alpha_k \rightarrow \alpha_{(k+1) \bmod p}$ , hence permuting the Voronoï cells  $V_k := \{z : \|\alpha_k - z\| < \|\alpha_j - z\|, \forall j \neq k\}$  by  $T_\omega(V_k) = V_{(k+1) \bmod p}$ . For any rotation-invariant Borel measure,  $\mu(V_k)$  is independent of  $k$ . This equipartition by cyclic symmetry is an elementary observation; it records that the basin bounds of Proposition 16.1 are distributed uniformly across the anchor indices.

## 20 No-cycle theorem for multi-start orbits

**Theorem 20.1** (No non-trivial cycles in the contraction disk). Let  $(x, p, \alpha)$  satisfy the standard hypotheses, and let  $z_0 \in \overline{B}(\alpha, R_\sigma/2)$ . If there exist integers  $0 \leq n < m$  with  $\sigma_{p,x}^n(z_0) = \sigma_{p,x}^m(z_0)$ , then  $\sigma_{p,x}^n(z_0) = \alpha$ .

*Proof.* The three-step construction of Part II supplies  $K_\sigma, R_\sigma$  with  $K_\sigma R_\sigma \leq 1/2$  and

$$|\sigma_{p,x}(z) - \alpha| \leq K_\sigma |z - \alpha|^2 \quad \text{on } \overline{B}(\alpha, R_\sigma).$$

For  $z \in \overline{B}(\alpha, R_\sigma/2)$ ,  $|z - \alpha| \leq R_\sigma/2$  gives

$$|\sigma_{p,x}(z) - \alpha| \leq K_\sigma \cdot (R_\sigma/2) \cdot |z - \alpha| \leq \frac{1}{4} |z - \alpha|.$$

Hence  $\sigma_{p,x}$  restricted to  $\overline{B}(\alpha, R_\sigma/2)$  is  $\frac{1}{4}$ -Lipschitz with  $\alpha$  as fixed point, and  $\sigma_{p,x}(\overline{B}(\alpha, R_\sigma/2)) \subset \overline{B}(\alpha, R_\sigma/2)$ .

Write  $w_j := \sigma_{p,x}^j(z_0)$ ; then  $|w_j - \alpha| \leq 4^{-j} |z_0 - \alpha|$  by induction. Suppose  $w_n = w_m$  with  $n < m$ . Applying the same bound from index  $n$ :  $|w_m - \alpha| \leq 4^{-(m-n)} |w_n - \alpha|$ . Combined with  $w_n = w_m$ :

$$|w_n - \alpha| \leq 4^{-(m-n)} |w_n - \alpha|,$$

and  $4^{-(m-n)} < 1$  forces  $|w_n - \alpha| = 0$ . □

**Corollary 20.1** (Acyclicity of seeded multi-start orbits). For every  $(x, p, \alpha, z)$  in the hypotheses of Theorem 16.1, the orbit of seed $_{\alpha, \varepsilon_{\text{seed}}}(z)$  under  $\sigma_{p,x}$  has no non-trivial cycles.

*Proof.* The seed lies in  $\overline{B}(\alpha, R_\sigma/2)$  (Definition 16.1), so Theorem 20.1 applies. □



## 21 Uniform log-log complexity over bounded $x$ -regimes

**Theorem 21.1** (Uniform log log complexity). For every compact  $K \subset (1, \infty)$  and every  $\varepsilon \in (0, 1)$ , there exists an integer  $N(K, \varepsilon)$  such that for every  $p \geq 2$ , every  $x \in K$ , and every starting point  $s_0 \in [1/2, 1]$ , the Steffensen–Pandrosion iteration reaches precision  $|v_n - x^{1/p}| \leq \varepsilon$  within at most  $N(K, \varepsilon)$  steps. Furthermore,  $N(K, \varepsilon) = \mathcal{O}(\log \log(1/\varepsilon))$  as  $\varepsilon \rightarrow 0$ , with constants depending only on  $K$ .

*Proof.* Set  $\lambda^*(K) := \sup_{x \in K} \lambda_\infty(x) < 1$  (Proposition 9.1(3):  $\lambda_\infty$  is continuous on  $(1, \infty)$  and strictly less than 1 pointwise, so its supremum on the compact  $K$  is strictly less than 1). Fix  $\delta := (1 - \lambda^*(K))/4 > 0$ .

By the Taylor bound (inequality (9)), for every  $p \geq p_0(K) := \lceil \ln(\sup K) \rceil + 1$  and every  $x \in K$ ,

$$\lambda_{p,x} \leq \lambda_\infty(x) + \frac{C(x)}{p} \leq \lambda^*(K) + \frac{C^*(K)}{p},$$

where  $C^*(K) := \sup_{x \in K} 3x(\ln x)^2/(x - 1) < \infty$ . Choose  $p_1(K) := \max(p_0(K), \lceil C^*(K)/\delta \rceil)$ , so that  $\lambda_{p,x} \leq \lambda^*(K) + \delta \leq 1 - 3\delta$  for every  $p \geq p_1(K)$  and  $x \in K$ .

For  $2 \leq p < p_1(K)$ ,  $\lambda_{p,x}$  is continuous on the compact  $\{2, 3, \dots, p_1(K) - 1\} \times K$  and strictly less than 1 pointwise (uniform monotonicity from Theorem 9.1); hence bounded away from 1 by some  $\lambda^{**}(K) < 1$ . Set  $\bar{\lambda}(K) := \max(1 - 3\delta, \lambda^{**}(K)) < 1$ , so that  $\lambda_{p,x} \leq \bar{\lambda}(K)$  for every  $p \geq 2$ ,  $x \in K$ .

The un-accelerated Pandrosion iteration then satisfies  $|s_n - s^*| \leq \bar{\lambda}(K)^n |s_0 - s^*|$ ; combined with the output-residual factor (Theorem 10.1) bounded on  $K$ , this gives linear convergence to  $\alpha$  with uniform rate. The Aitken–Steffensen upgrade (Theorem 13.1) turns this into quadratic convergence with constant  $K_S(K) := \sup_{(p,x) \in \{p \geq 2\} \times K} |h''(s^*)/(2(1 - \lambda_{p,x}))|$ , uniform on  $K$  by continuity plus  $\lambda_{p,x} \leq \bar{\lambda}(K)$ .

Combining:  $n_1 := \lceil C_1(K) \log(1/R_\sigma(K)) \rceil$  linear iterations drive any  $s_0 \in [1/2, 1]$  into the quadratic-contraction disk of radius  $R_\sigma(K) := \inf_{(p,x)} R_\sigma(x, p, \alpha) > 0$ ; then  $n_2 := N_{\text{eff}}(K_S(K), R_\sigma(K), \varepsilon)$  Aitken–Steffensen iterations (Theorem 16.2) reach precision  $\varepsilon$ . The total  $N(K, \varepsilon) = n_1 + n_2 = \mathcal{O}_K(\log \log(1/\varepsilon))$  is uniform in  $p \geq 2$  and  $x \in K$ .  $\square$

This is the uniform version of the per-regime benchmark of §14.3: the observed  $\approx 3$ –4 additional iterations when moving from 50 to 500 digits, *across all tested*  $(p, x)$ , is a quantitative instance of Theorem 21.1.

## 22 Conclusion

The central tension of this paper is between Pandrosion’s ancient geometric construction — an order-one iteration requiring only a ruler and parallels — and modern acceleration theory. Once the universal iteration  $s_{n+1} = 1 - (x - 1)/(x S_p(s_n))$  is written down, three ingredients suffice to make it competitive on  $\mathbb{R}$ : the closed-form contraction ratio  $\lambda_{p,x}$  (Theorem 5.1), the preconditionings of Propositions 12.1 and 14.1, and the Aitken  $\Delta^2$  upgrade (Theorem 13.1). The ratio  $\lambda_{p,x} = h'(s^*)$  is monotone in both  $p$  and  $x$  uniformly (Theorem 9.1, via the closed-form identities  $P_p = p \cdot D'_p$  and  $B_p = \sum_{j=1}^p j \alpha^{j-1}$ ), has an explicit large- $p$  limit  $1 - \ln(x)/(x - 1)$  (Proposition 9.1), and admits a clean threshold analysis for  $x < 1$  (Proposition 11.1).

On  $\mathbb{C}$ , the extension via the target-dependent seed of Definition 16.1 yields a universal multi-start convergence theorem (Theorem 16.1) with a closed-form  $N_{\text{eff}} \sim \log_2 \log(1/\varepsilon)$  iteration count (Theorem 16.2), together with Borel-measurability and a.e.-continuity of the induced solution map (Theorems 17.1–17.3). All of these results are formally verified in the Lean 4 corpus described in Section 18.

The paper does not claim progress on classical open problems of polynomial root-finding, and does not claim a general algorithmic advantage over Newton or its modern variants. Its

main deliverable is a concrete derivative-free root-finding scheme with closed-form convergence analysis and complete formal verification for the monomial family  $z^p = x$ , connecting a ruler-and-parallel construction from fourth-century Alexandria to contemporary formal-methods practice.

**Perspectives.** We see four concrete, tractable extensions as immediate next steps:

1. **Benchmark against Schröder / König families of order  $k \geq 3$**  at multi-precision, to quantify how the factor-of-3 cubic gap of Halley over Newton compares against purely algebraic higher-order iterations (the König family is derivative-free in spirit and admits closed-form constants).
2. **Extend to complex  $x$**  (not just complex anchor  $\alpha$ ): the current Part II assumes  $x \in \mathbb{C}^*$  in the statements but the analysis is principally real for the scaling step. A fully complex scaling, via  $\beta \in \mathbb{Z}[i]$  close to  $X^{1/p}$  in the Gaussian integers, is a natural extension.
3. **Formalize the non-asymptotic bounds of Section 10 in Lean**, closing the gap between the analytic quantity  $\Lambda$  and its in-Lean counterpart. This is a routine but useful extension of `UniformContractionRate.lean`.
4. **Matching upper bound on the basin** (Remark 16.6 pointed the way): show that the full basin  $\Omega_k$  equals  $\bigcup_{m \geq 0} \sigma_{p,x}^{-m}(\overline{B}(\gamma_k, R_\sigma))$  up to a Lebesgue-null set, yielding a matching upper bound on  $\text{vol}(\Omega_k)$ . This stops short of requiring Fatou–Julia machinery.

A Fatou–Julia formalization in Mathlib — which would enable a proper in-Lean proof of McMullen’s 1987 theorem — is a much larger project and is mentioned here only for completeness, not as an immediate extension.

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